

Introduction to Signals

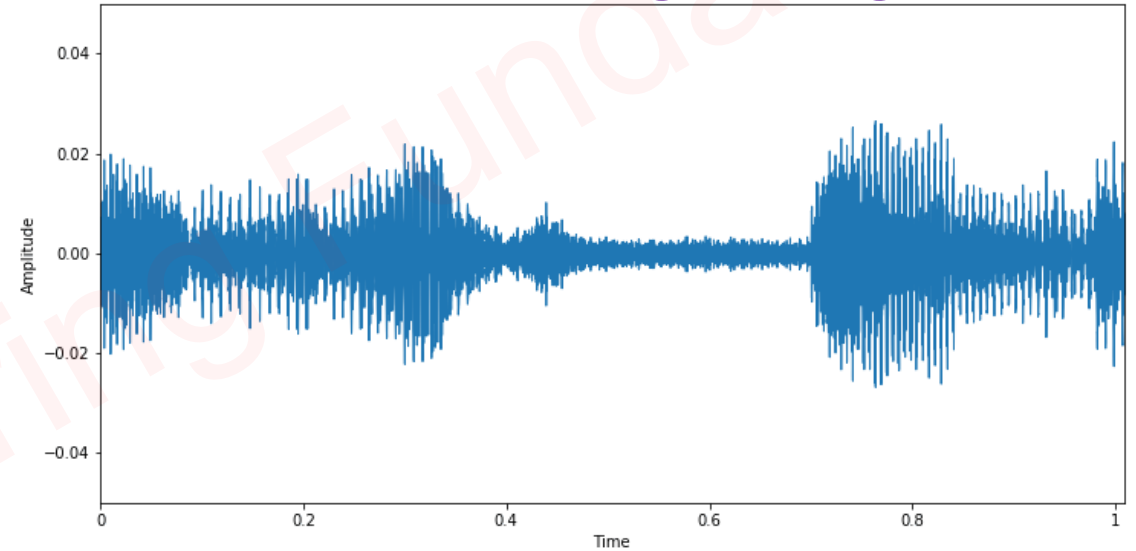
Outlines

- Basics of Signals
- Representation of Signals
- Applications of Signals

Engineering Funda

Basics of Signals

- ❑ A Signals can be any Physical quantity that varies with time, space, or independent variable.
- ❑ Signals are used to convey information in different branches of Engineering.
- ❑ Examples:
 - Stock Market Index
 - Speech Signal
 - Heart Rate

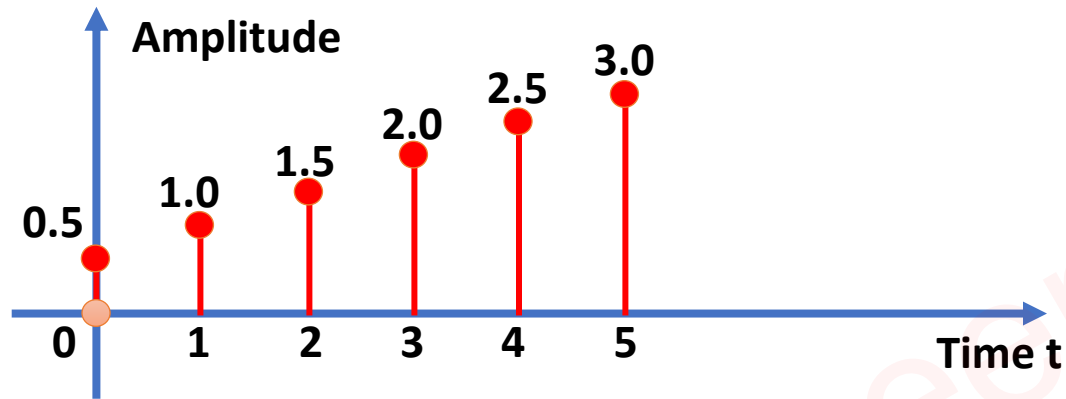


Representation of Signals

□ Functional Representation:

$$\Rightarrow x(n) = \frac{n+1}{2}, \{0 \leq n \leq 5\}$$

□ Graphical Representation:



□ Tabular Representation:

N	0	1	2	3	4	5
$x(n]$	0.5	1.0	1.5	2.0	2.5	3.0

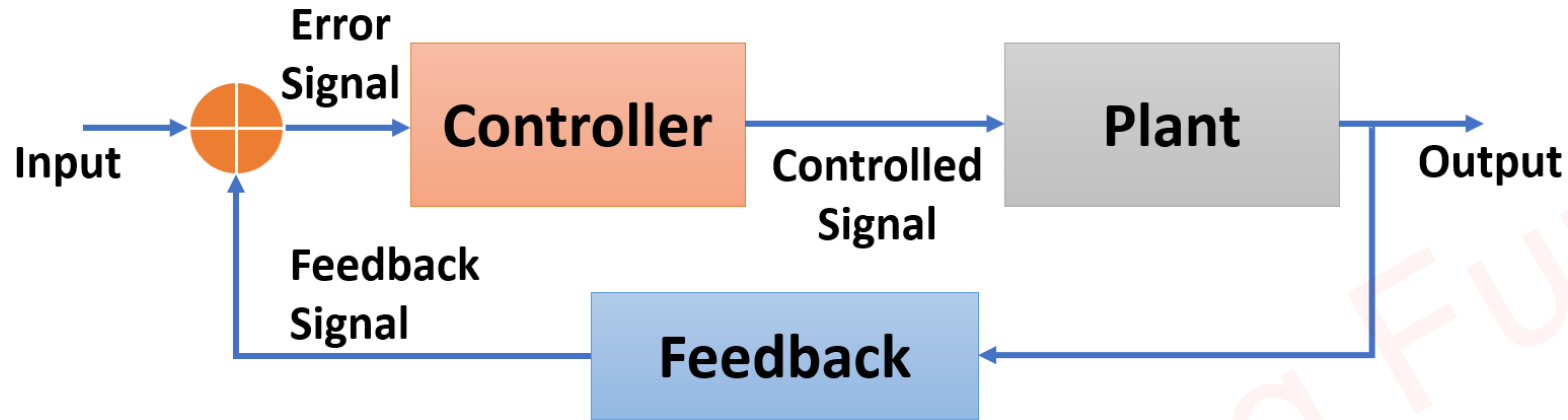
□ Sequence Representation:

$$\Rightarrow x(n] = \{0.5, \quad 1.0, \quad 1.5, \quad 2.0, \quad 2.5, \quad 3.0\}$$

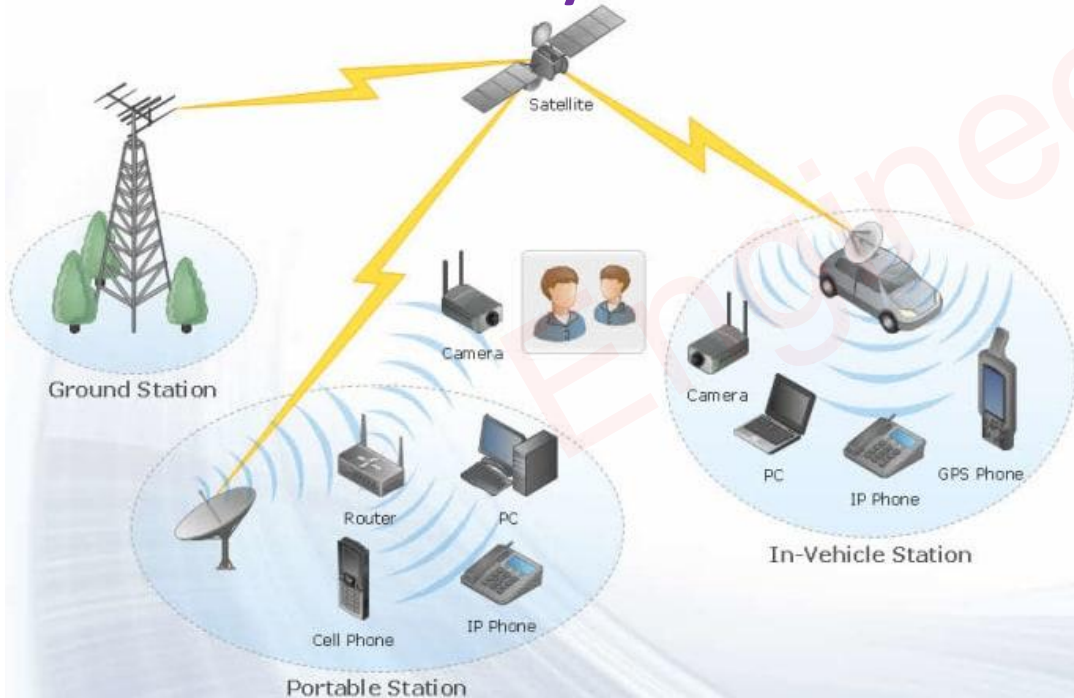
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Applications of Signals

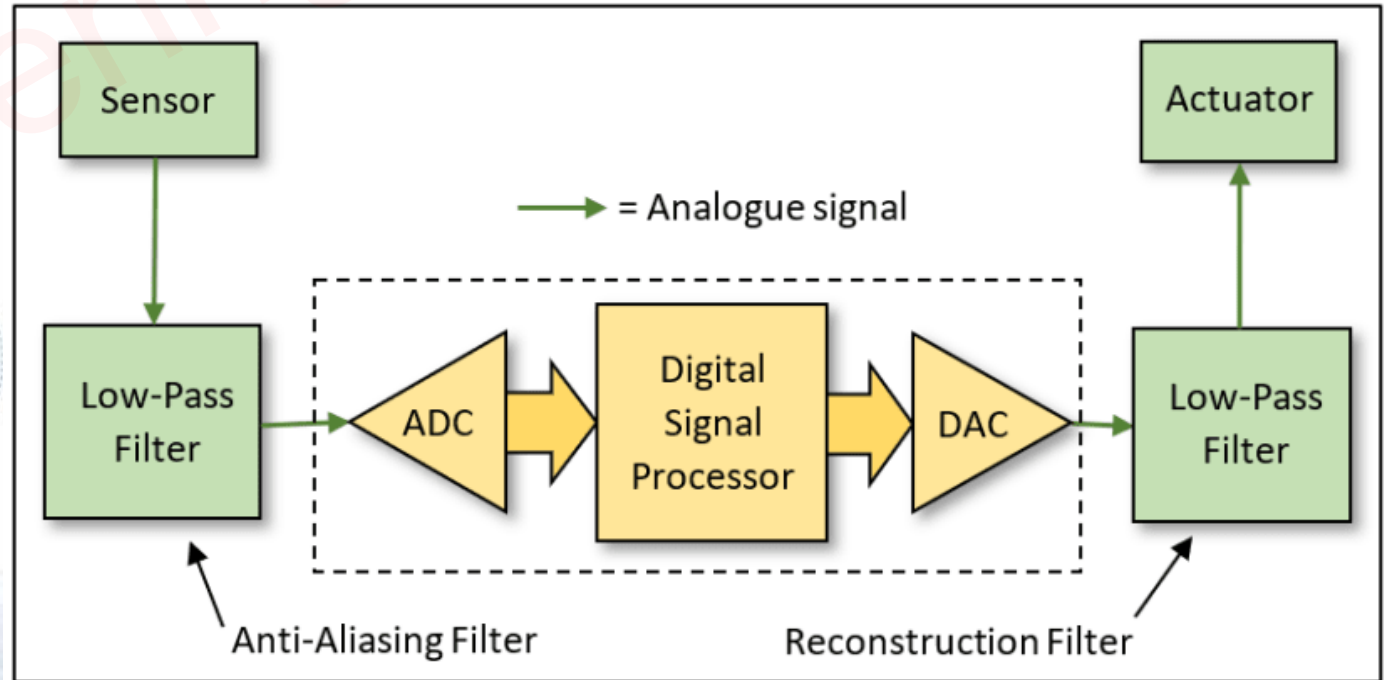
❑ Control Systems:



❑ Communication Systems:



❑ Signal Processing:



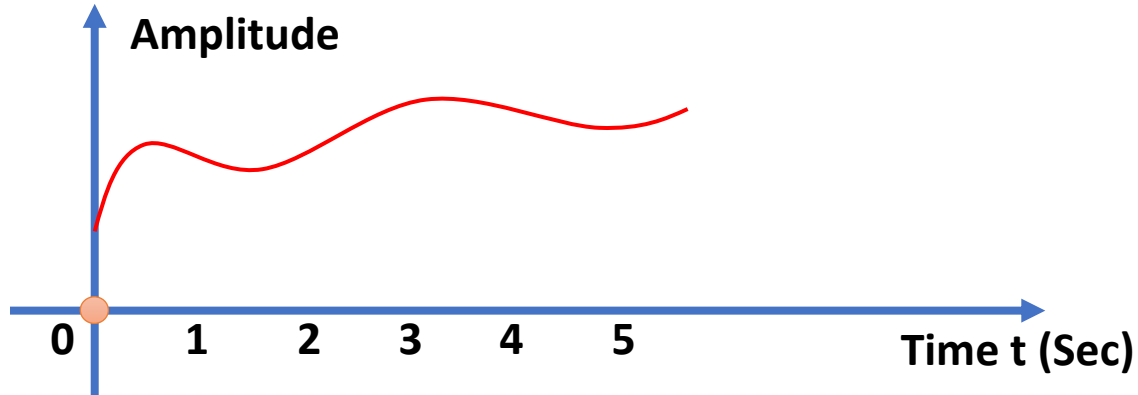
Classification of Signals

Signals can be Classified in following ways:

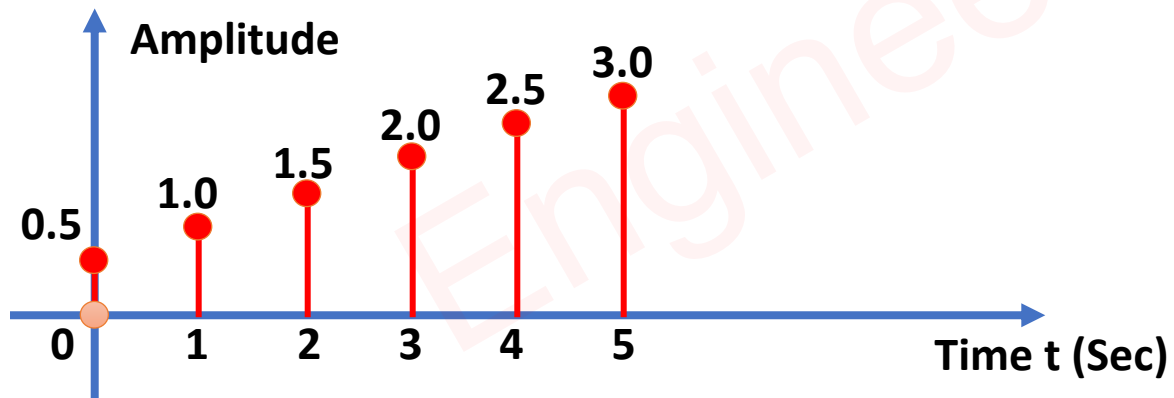
- Continuous Time and Discrete Time Signals
- Analog and Digital Signals
- Deterministic and Random Signals
- Even and Odd Signals
- Periodic and Aperiodic Signals
- Energy and Power Signals

Continuous Time and Discrete Time Signals

□ A signal of Continuous amplitude and time is known as a Continuous Time signal.

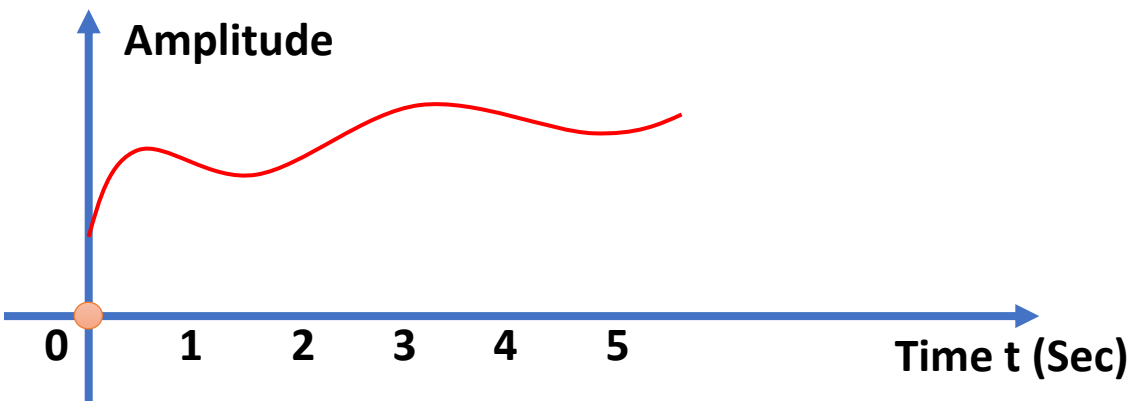


□ A signal that has Amplitude at Discrete-time intervals is known as a Discrete Time signal.

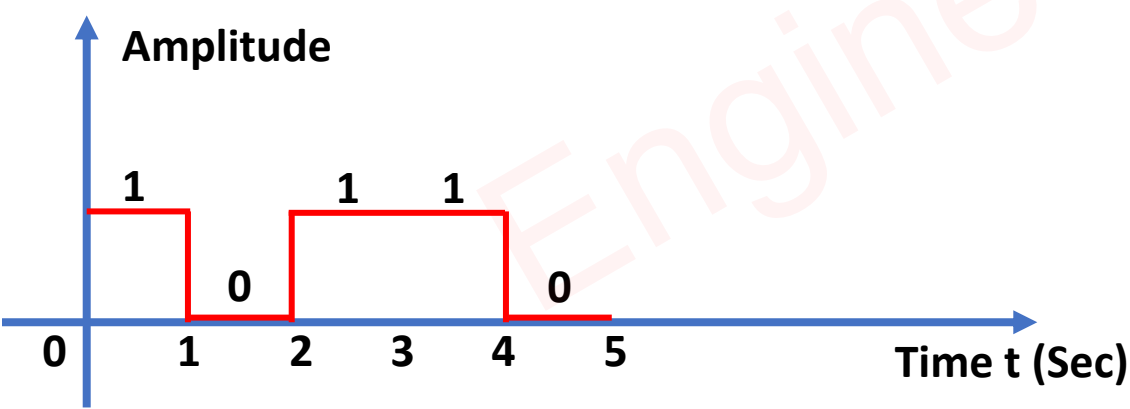


Analog and Digital Signals

❑ Analog signals are continuous signals that vary smoothly over time and can take any value within a range.



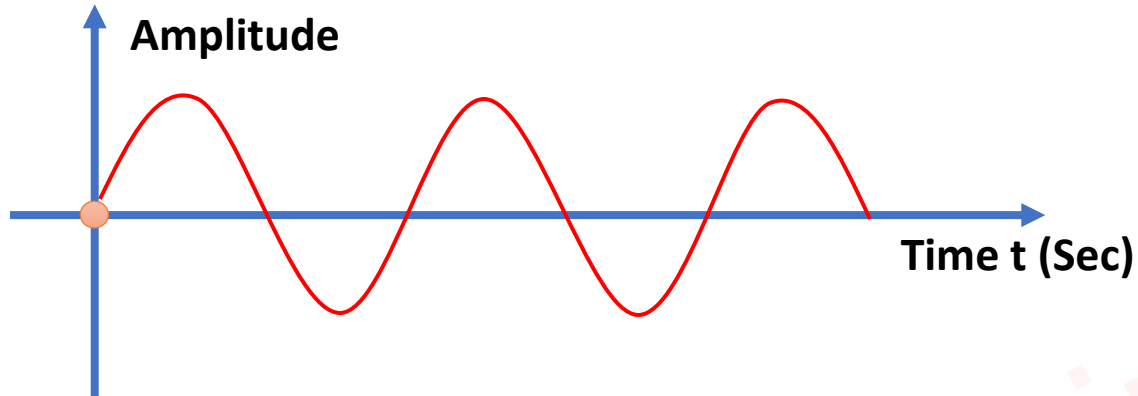
❑ Digital signals are Discrete in Nature (Discrete in time and Amplitude) and have binary values for 0's and 1's.



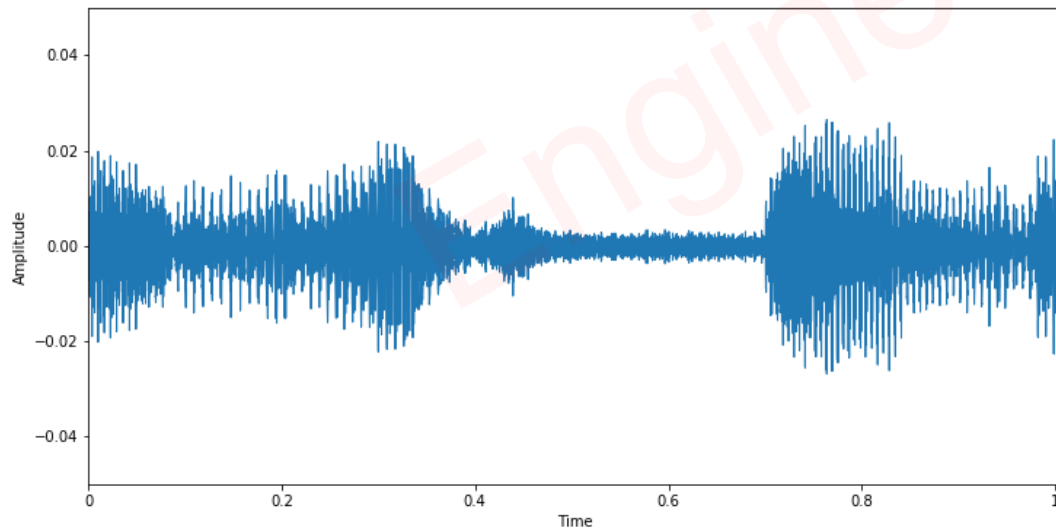
Features	Analog Signal	Digital Signal
Nature	Continuous	Discrete
Representation	Waveforms & Functions	Binary, Hex, Octal
Noise Susceptibility	High	Low
Bandwidth	Low	High
Examples	Speech Signals	CD, Hard Disk

Deterministic and Random Signals

- ❑ Deterministic signals can be defined or Predicted using Mathematical Functions.
- ❑ It can be Determined from present and past data.



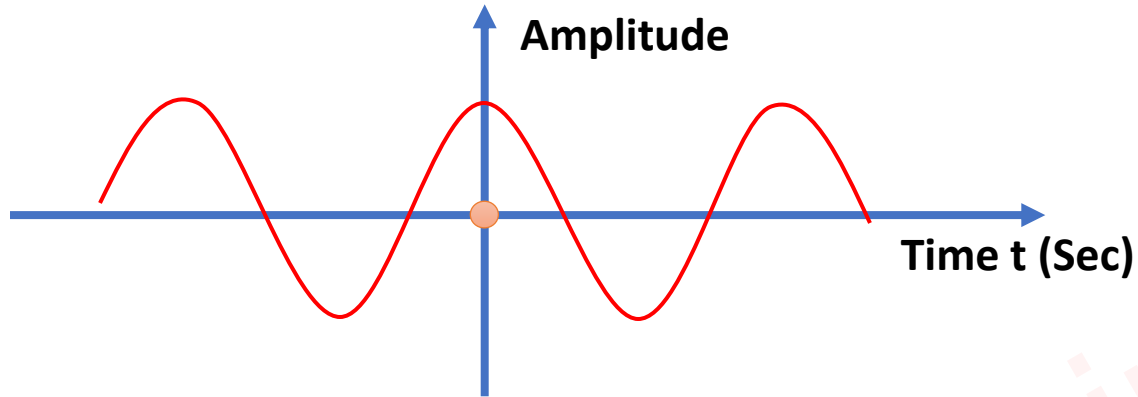
- ❑ Random signals Can not be defined or Predicted using Mathematical Functions.
- ❑ It exhibits Uncertainty and Randomness in its Behaviour.



Even and Odd Signals

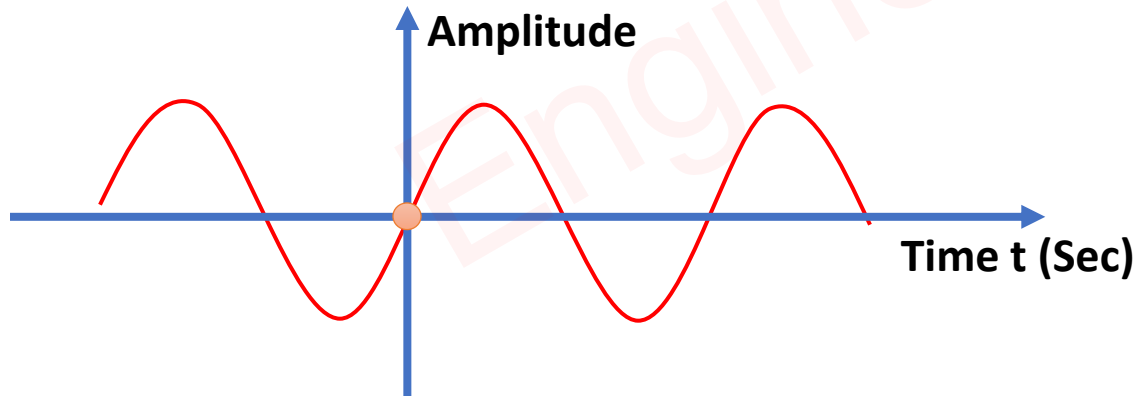
□ Even Signal follows the following Condition:

$$\Rightarrow x(n) = x(-n)$$



□ Odd Signal follows the following Condition:

$$\Rightarrow x(n) = -x(-n)$$



□ Important Notes:

- A sum of two or more even functions is an even function.
- A sum of two or more Odd functions is an Odd function.
- The product of two or more even functions is an even function.
- The product of even numbers of odd functions is an even function.
- The product of Odd numbers of odd functions is an Odd function.

□ Even and Odd Components of Signal:

$$\Rightarrow x(t) = x_{\text{even}}(t) + x_{\text{odd}}(t)$$

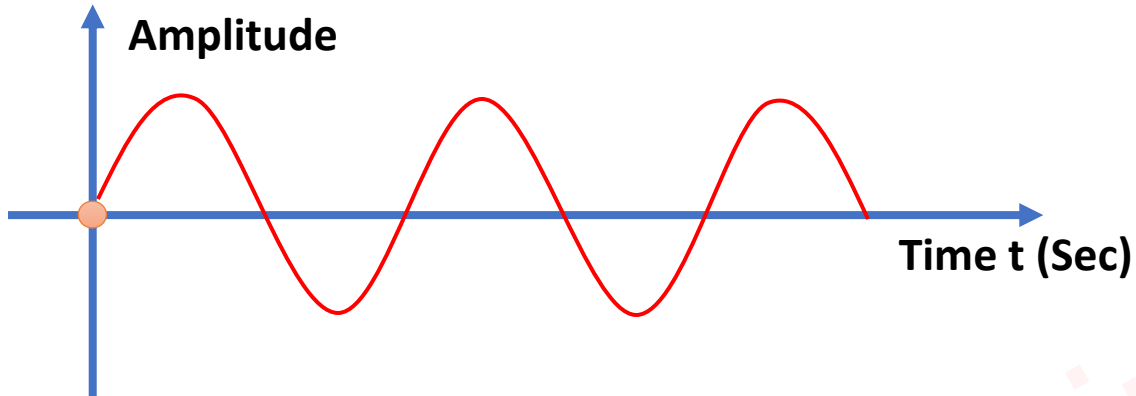
$$\Rightarrow x_{\text{even}}(t) = \frac{x(t) + x(-t)}{2}$$

$$\Rightarrow x_{\text{odd}}(t) = \frac{x(t) - x(-t)}{2}$$

Periodic and Aperiodic Signals

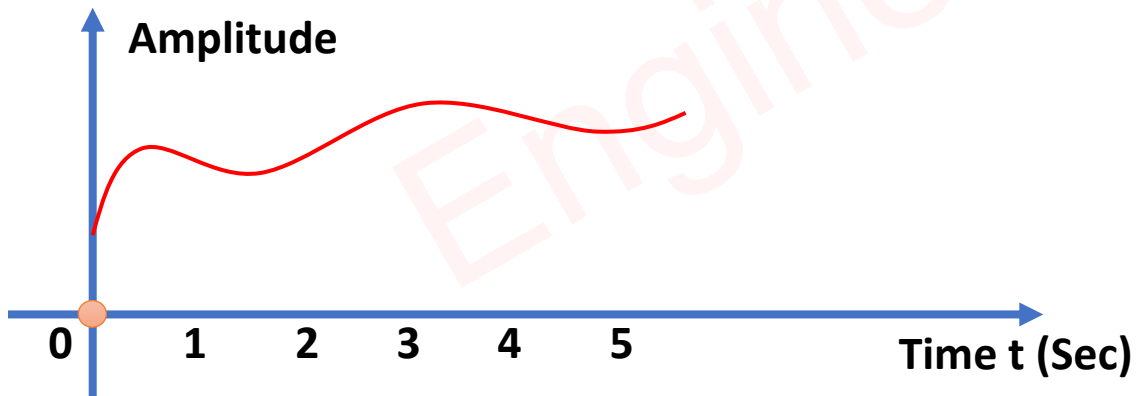
□ If the signal repeats itself after a fixed period (T), it is called Periodic Signal.

$$\Rightarrow x(t) = x(t + T)$$



□ If the signal does not repeat itself after any finite period (T), it is called Aperiodic Signal.

$$\Rightarrow x(t) \neq x(t + T)$$



Energy and Power Signals

- ❑ If the signal is an Energy Signal, it can not be a Power Signal or vice versa, but it is possible that the signal may not be an Energy and Power Signal.
- ❑ If the Signal has Finite Energy and Zero Power then the signal is an Energy Signal $x(t)$.

$$\Rightarrow E = \int_{-\infty}^{+\infty} x(t)^2 dt$$

- ❑ If the Signal has Finite Power and Infinite Energy then the signal is a Power Signal $x(t)$.
Almost all the Periodic Signals are Power Signals.

$$\Rightarrow P = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{+T/2} x(t)^2 dt$$

Features	Energy Signal	Power Signal
Energy	Finite	Infinite
Power	Zero	Finite
Examples	Pulse	Periodic Signals

Standard Signals

Some Standard Signals are as follows:

- Unit Impulse Signal
- Unit Step Signal
- Unit Ramp Signal
- Unit Parabolic Signal
- Signum Function
- Rectangular Pulse
- Sinc Function & Sampling Function

Unit Impulse Signal $\delta(t)$ and $\delta(n)$

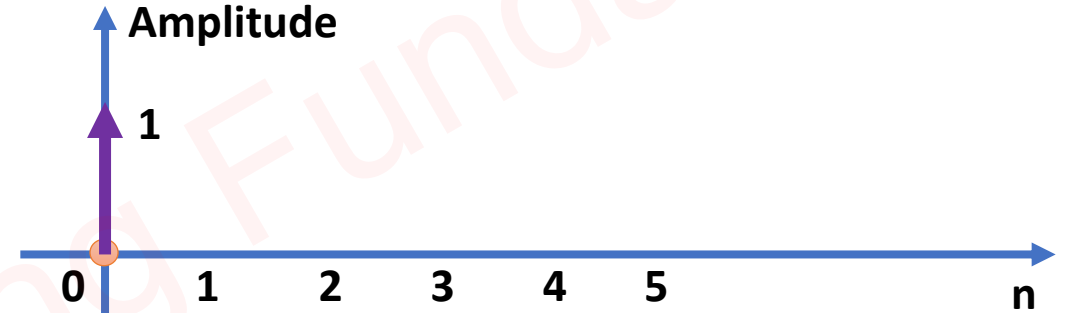
□ Function of Unit Impulse Signal $\delta(t)$ and $\delta(n)$:

$$\Rightarrow \delta(t) = 1, \quad t = 0$$

$$\Rightarrow \delta(t) = 0, \quad t \neq 0$$

$$\Rightarrow \delta(n) = 1, \quad n = 0$$

$$\Rightarrow \delta(n) = 0, \quad n \neq 0$$



□ Impulse Signal in Laplace Domain = 1

□ Properties of Unit Impulse Signal

- $\int_{-\infty}^{\infty} \delta(t) dt = 1$
- $\delta(n - k) = 1$ at $n = k$
- $\delta(n) = u(n) - u(n - 1)$
- $f(t)\delta(t) = f(0)$ as Impulse is there at $t = 0$.
- $f(t)\delta(t - t_0) = f(t_0)$ as Impulse is there at $t = t_0$.
- $\delta(kt) = \frac{1}{|k|} \delta(t)$
- $\delta(-t) = \delta(t)$ Unit Impulse signal is even function.

Unit Step Signal $u(t)$ and $u(n)$

□ Function of Unit Step Signal $u(t)$ and $u(n)$:

$$\Rightarrow u(t) = 1, \quad t \geq 0$$

$$\Rightarrow u(t) = 0, \quad t < 0$$

$$\Rightarrow u(n) = 1, \quad n \geq 0$$

$$\Rightarrow u(n) = 0, \quad n < 0$$



□ Unit Step Signal in Laplace Domain = $1/S$

□ Properties of Unit Step Signal

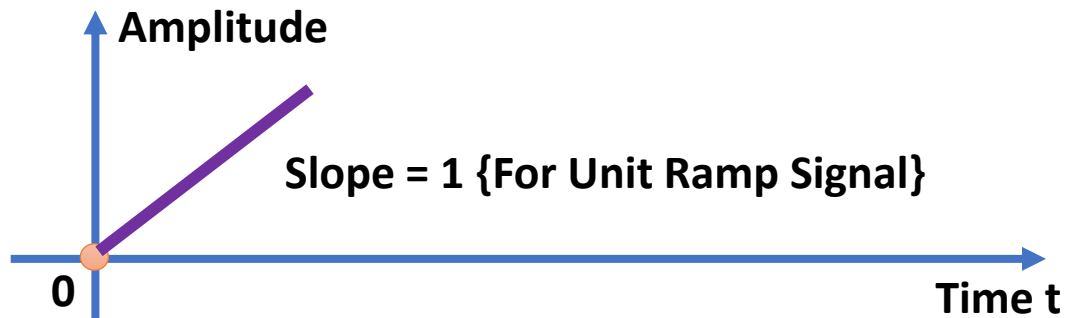
- $(u(t))^n = u(t)$
- $(u(t - t_0))^n = u(t - t_0)$
- $u(at) = u(t)$
- Example $\Rightarrow u(at - t_0) = u\left(a\left(t - \frac{t_0}{a}\right)\right) = u\left(t - \frac{t_0}{a}\right)$
- Example $\Rightarrow u(2t - 4) = u(2(t - 2)) = u(t - 2)$

Unit Ramp Signal $r(t)$ and $r(n)$

□ Function of Unit Ramp Signal $r(t)$ and $r(n)$:

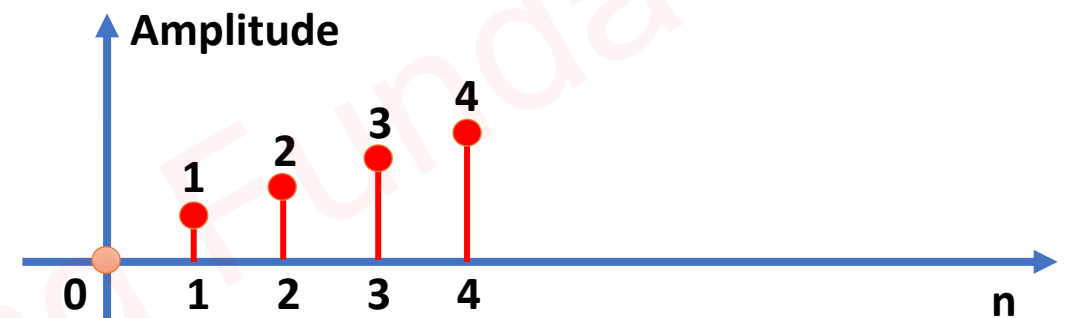
$$\Rightarrow r(t) = t, \quad t \geq 0$$

$$\Rightarrow r(t) = 0, \quad t < 0$$



$$\Rightarrow r(n) = n, \quad n \geq 0$$

$$\Rightarrow r(n) = 0, \quad n < 0$$



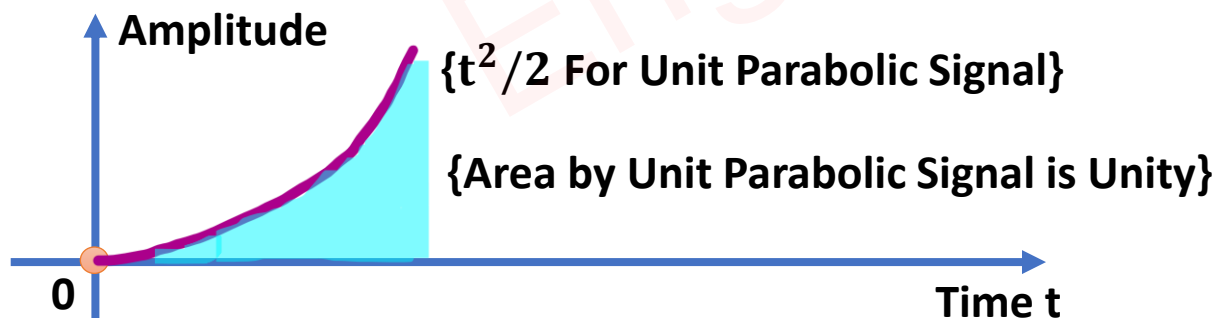
□ Ramp Signal in Laplace Domain = $1/S^2$

Unit Parabolic Signal $x(t)$ and $x(n)$

□ Function of Unit Parabolic Signal $x(t)$ and $x(n)$:

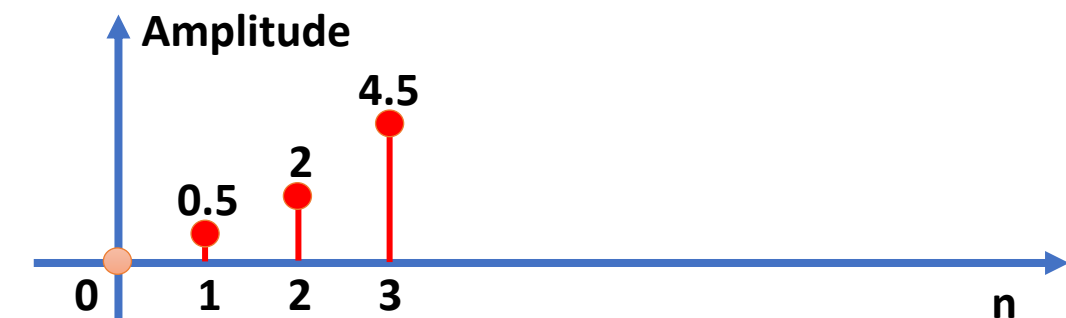
$$\Rightarrow x(t) = t^2/2, \quad t \geq 0$$

$$\Rightarrow x(t) = 0, \quad t < 0$$



$$\Rightarrow x(n) = n^2/2, \quad n \geq 0$$

$$\Rightarrow x(n) = 0, \quad n < 0$$



Relationship of Standard Test Signals

Impulse Signal $\delta(t)$	Step Signal $u(t)$	Ramp Signal $r(t)$	Parabolic Signal $x(t)$
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Integration with respect to time

Differentiation with respect to time

$$\Rightarrow \int \delta(t) dt = u(t)$$

$$\Rightarrow \int u(t) dt = r(t)$$

$$\Rightarrow \int r(t) dt = x(t)$$

$$\Rightarrow \frac{dx(t)}{dt} = r(t)$$

$$\Rightarrow \frac{dr(t)}{dt} = u(t)$$

$$\Rightarrow \frac{du(t)}{dt} = \delta(t)$$

Signum Function $S_{gn}(t)$ and $S_{gn}(n)$

□ Function of Signum Function $S_{gn}(t)$ and $S_{gn}(n)$:

$$\Rightarrow S_{gn}(t) = 1, \quad t > 0$$

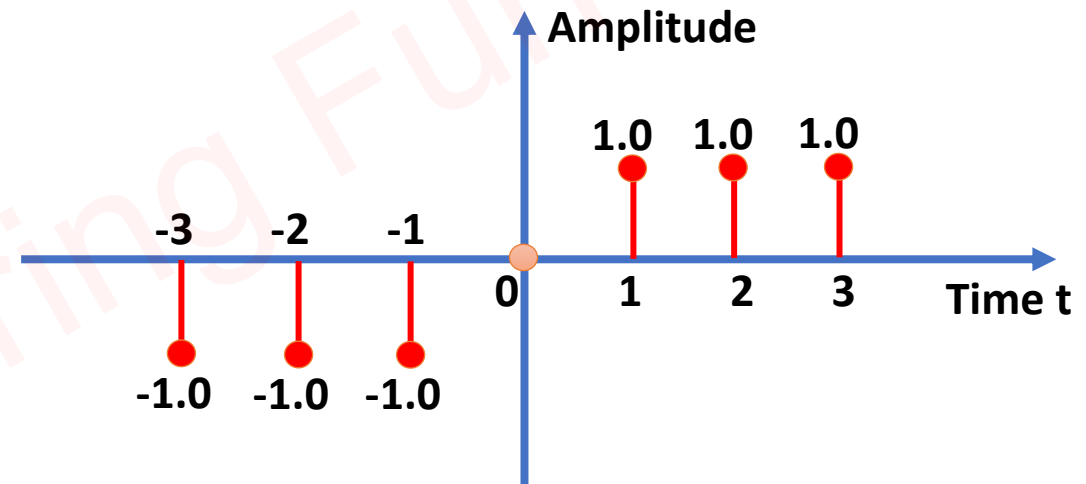
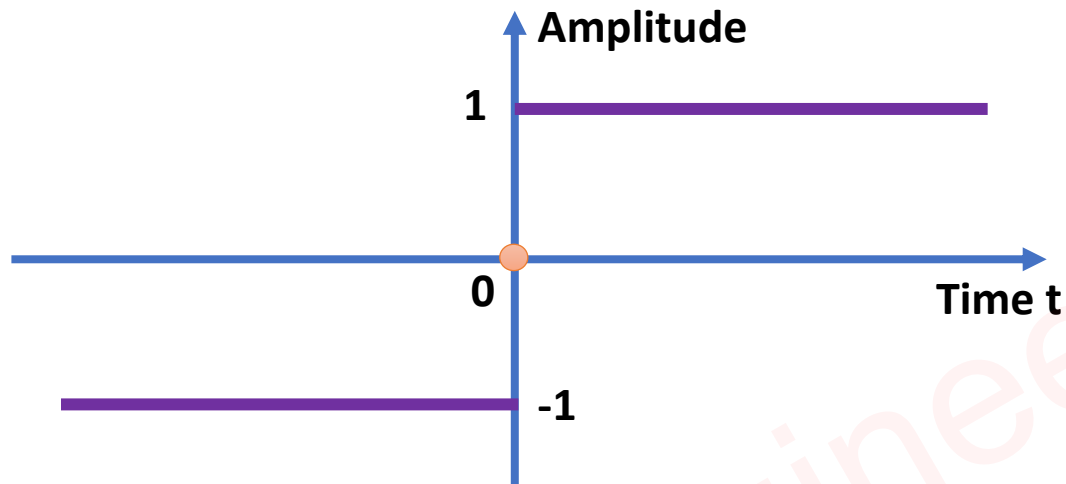
$$\Rightarrow S_{gn}(t) = 0, \quad t = 0$$

$$\Rightarrow S_{gn}(t) = -1, \quad t < 0$$

$$\Rightarrow S_{gn}(n) = 1, \quad n > 0$$

$$\Rightarrow S_{gn}(n) = 0, \quad n = 0$$

$$\Rightarrow S_{gn}(n) = -1, \quad n < 0$$



□ Relationship of Signum Function $S_{gn}(t)$ with Unit Step Signal $u(t)$:

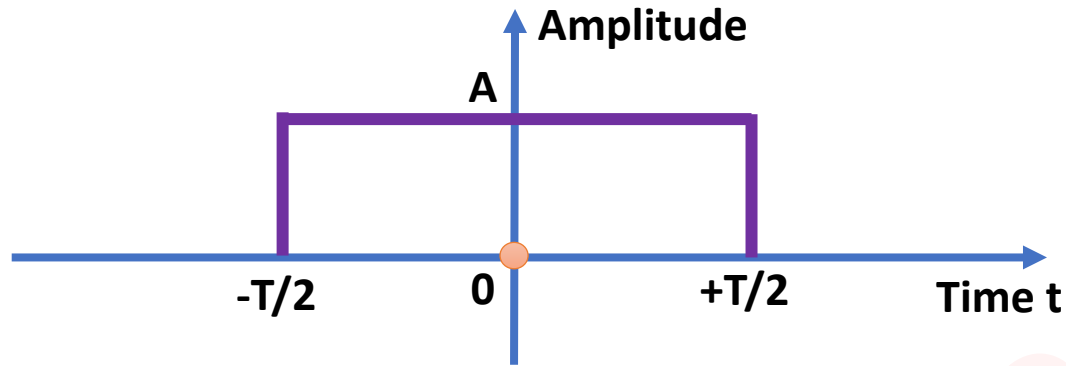
$$\Rightarrow S_{gn}(t) = 2u(t) - 1$$

Rectangular Pulse $x(t)$

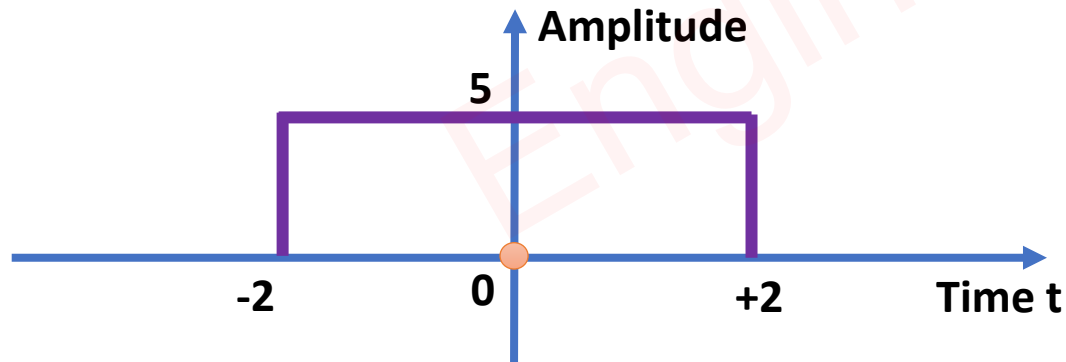
□ Function of Rectangular Pulse $x(t)$:

$$\Rightarrow x(t) = A \operatorname{rect}\left(\frac{t}{T}\right)$$

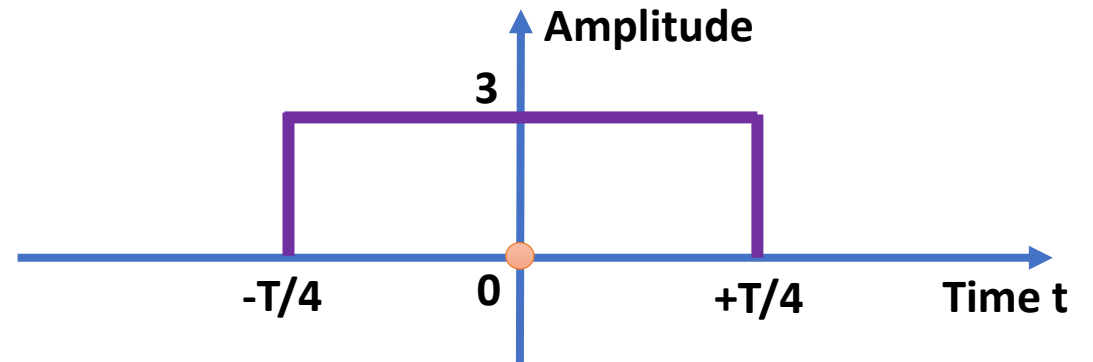
Where, A is Amplitude of Rectangular Pulse and T is Width of Pulse.



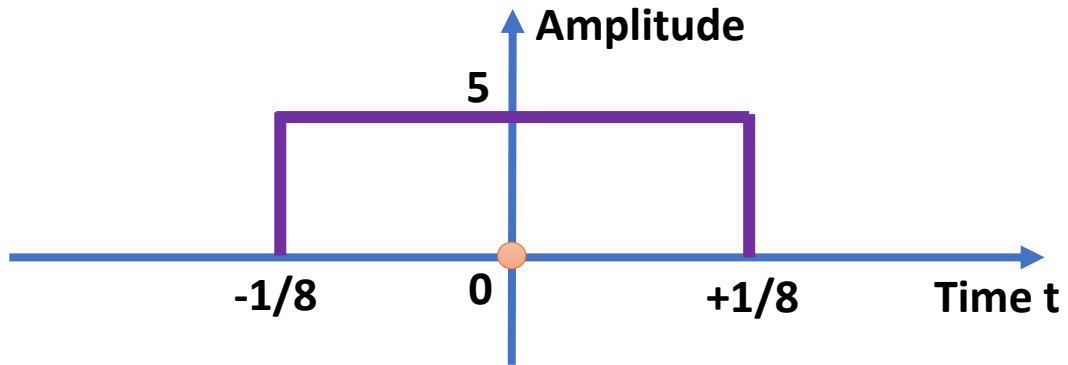
- Example: $x(t) = 5 \operatorname{rect}\left(\frac{t}{4}\right)$



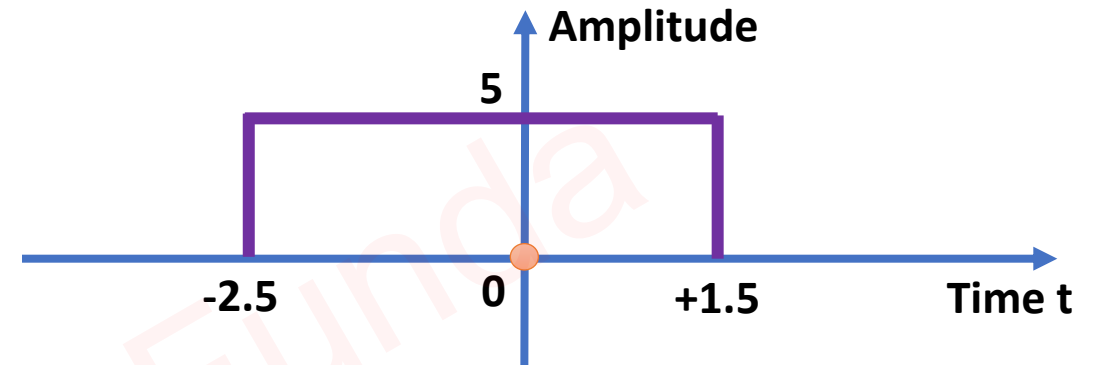
- Example: $x(t) = 3 \operatorname{rect}\left(\frac{2t}{T}\right)$



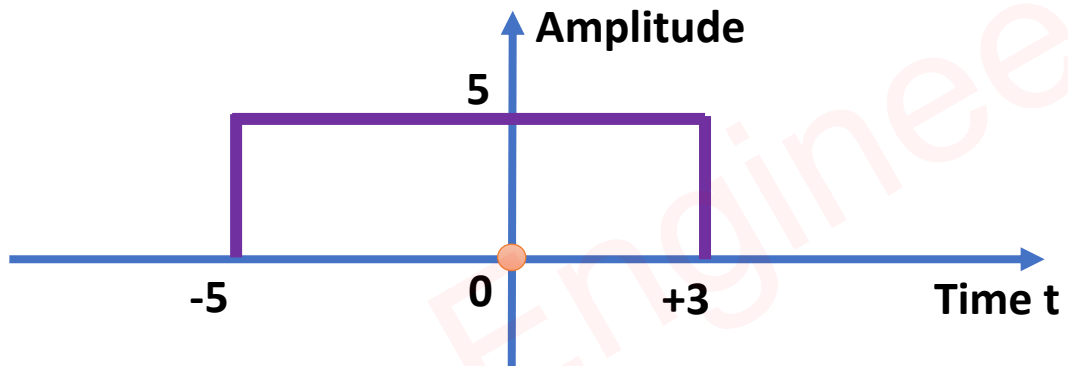
- Example: $x(t) = 5 \text{ rect}(4t)$



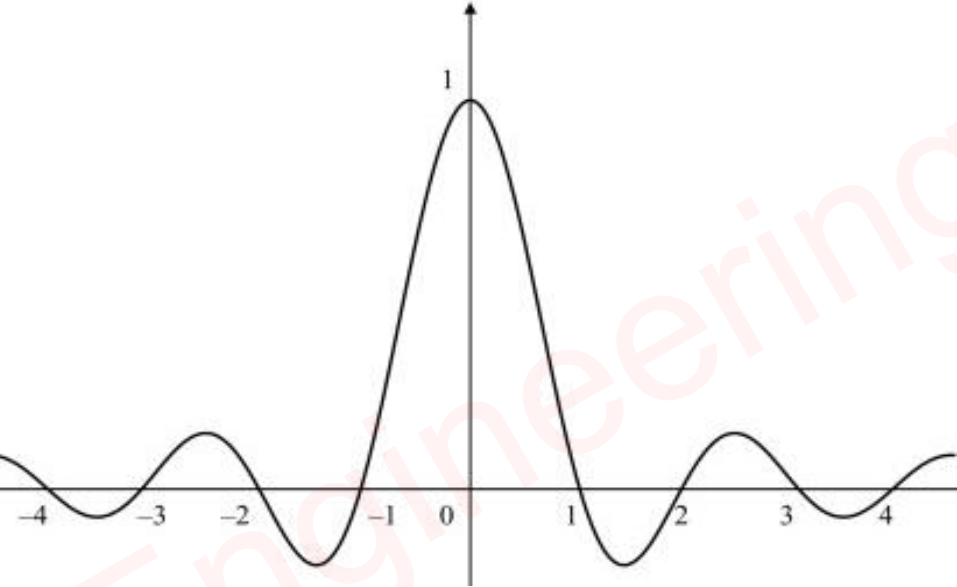
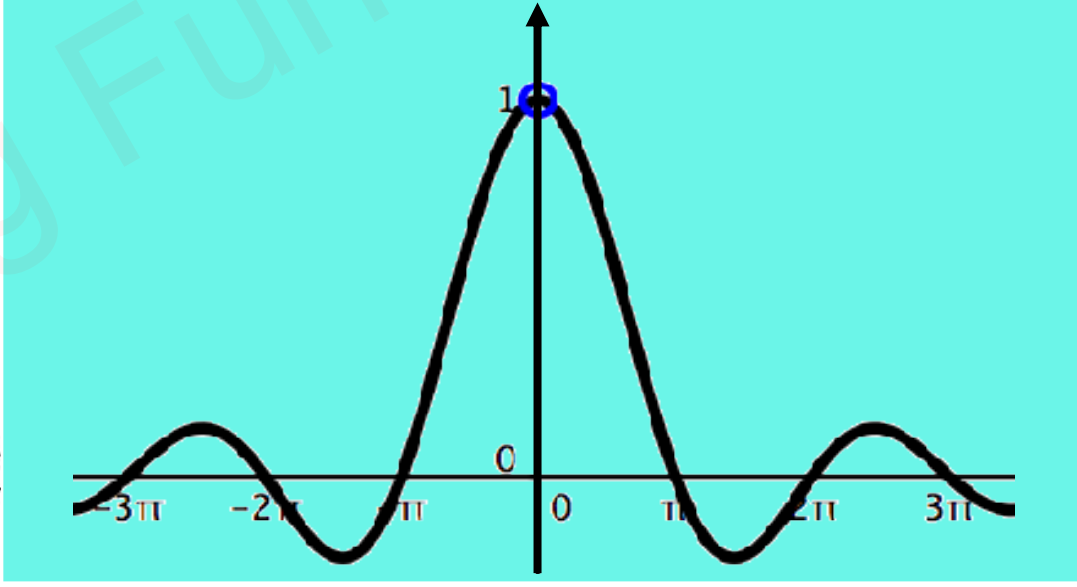
- Example: $x(t) = 5 \text{ rect}((2t + 1)/8)$



- Example: $x(t) = 5 \text{ rect}((t + 1)/8)$



Sinc Function and Sampling Function

Parameters	Sinc Function	Sampling Function
Notation	❖ Sinc (x)	❖ Sa (x)
Normalization	❖ It is Normalized Function	❖ It is not Normalized Function
Function	❖ $\text{Sinc}(x) = \frac{\sin \pi x}{\pi x}$	❖ $\text{Sa}(x) = \frac{\sin x}{x}$
Plot		

Operations on Signals

Following Operations can be done on Signals:

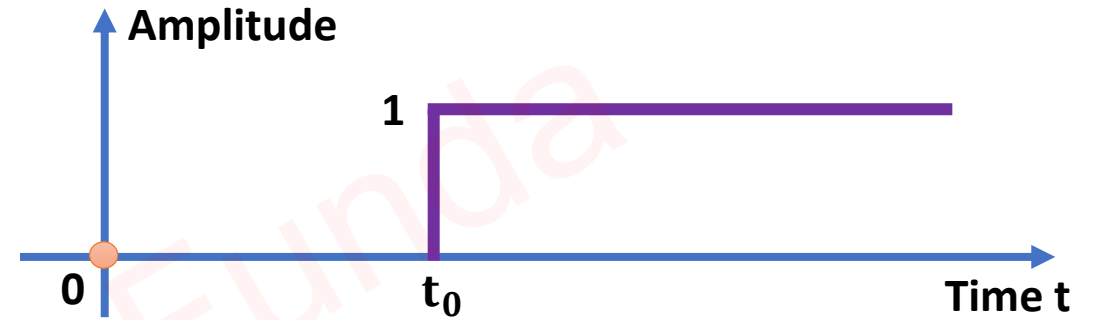
- Time Shifting Operation
- Time Folding Operation
- Time Folding and Shifting
- Time Scaling Operation
- Amplitude Scaling Operation
- Arithmetic Operation

Time Shifting Operation

❑ Delayed Signal $x(t - t_0)$ and $x(n - k)$:



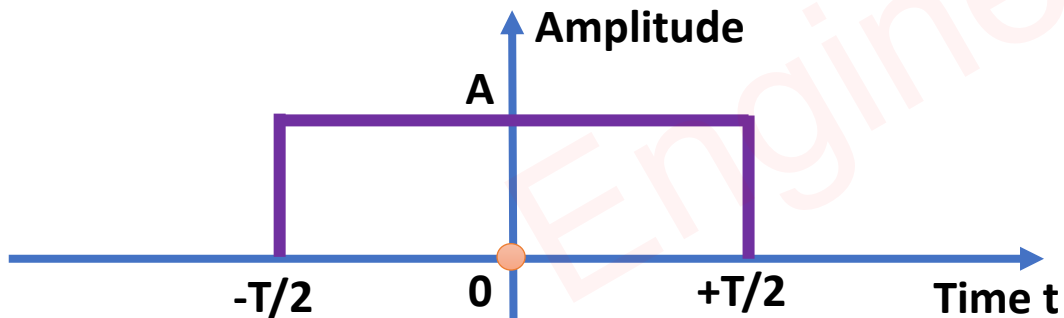
Delayed



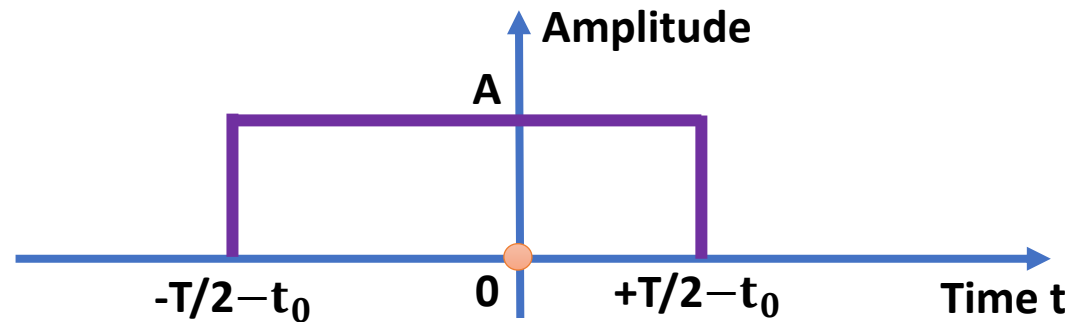
$$\Rightarrow x(n) = \{0.5, 1.0, 1.5, 2.0, 2.5, 3.0\}$$



❑ Advanced Signal $x(t + t_0)$ and $x(n + k)$:



Advanced



$$\Rightarrow x(n) = \{0.5, 1.0, 1.5, 2.0, 2.5, 3.0\}$$



$$\Rightarrow x(n + 2) = \{0.5, 1.0, 1.5, 2.0, 2.5, 3.0\}$$

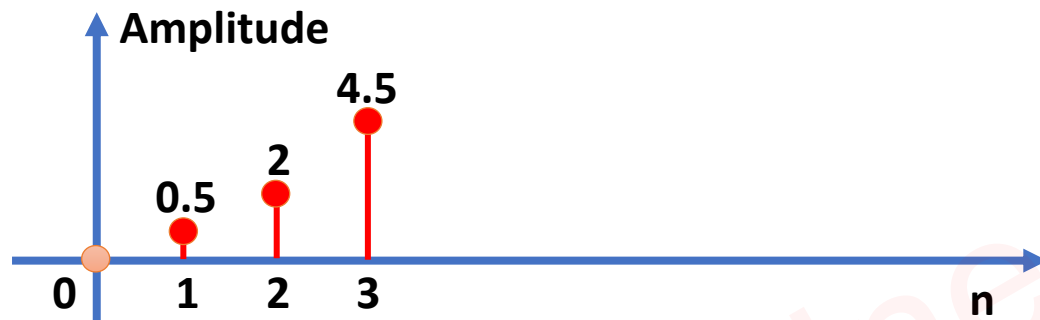


Time Folding Operation

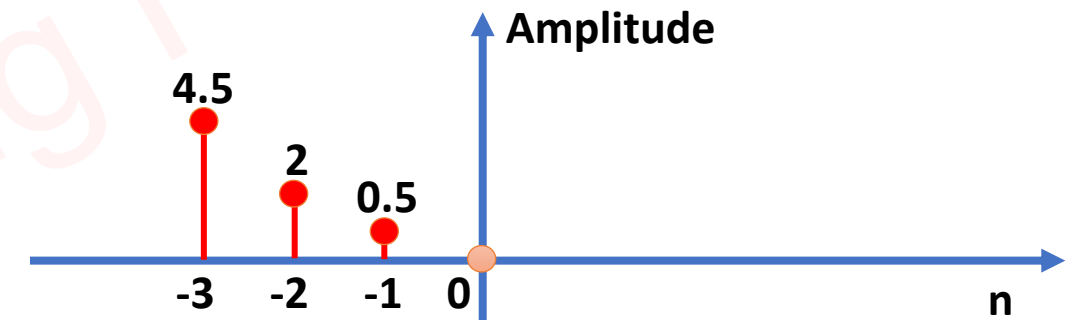
❑ Folded Signal $x(-t)$ and $x(-n)$:



Folding



Folding



$$\Rightarrow x(n) = \{0.5, 1.0, 1.5, 2.0, 2.5, 3.0\}$$



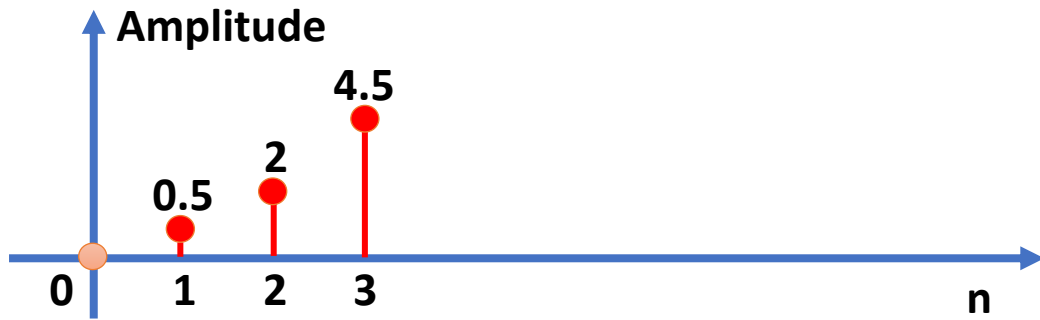
Folding

$$\Rightarrow x(-n) = \{3.0, 2.5, 2.0, 1.5, 1.0, 0.5\}$$

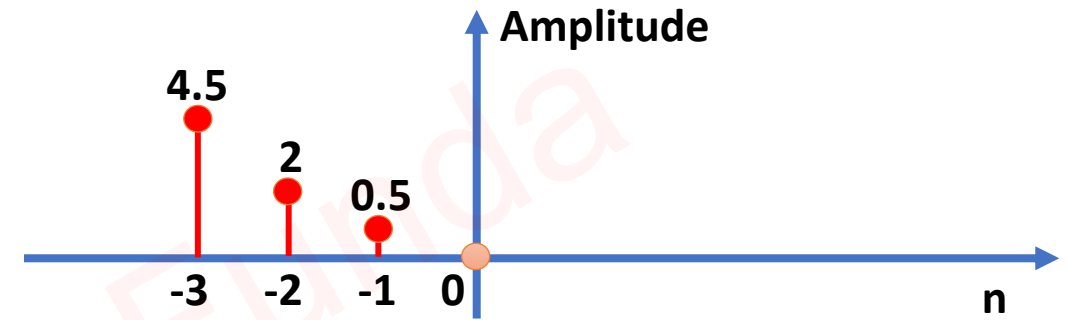


Time Folding and Shifting Operation

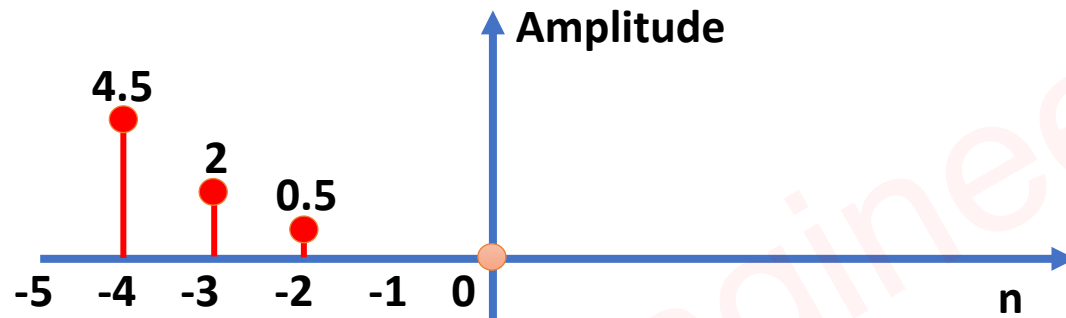
$$\Rightarrow x(n)$$



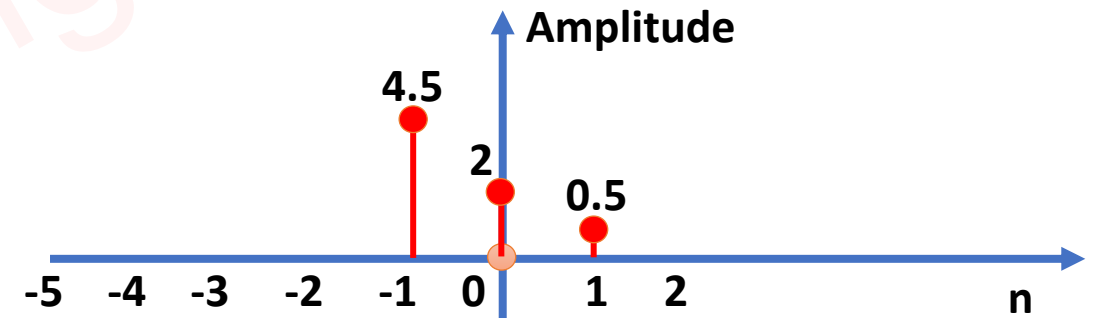
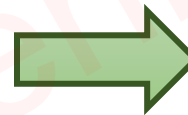
$$\Rightarrow x(-n)$$



$$\Rightarrow x(-n - 2)$$

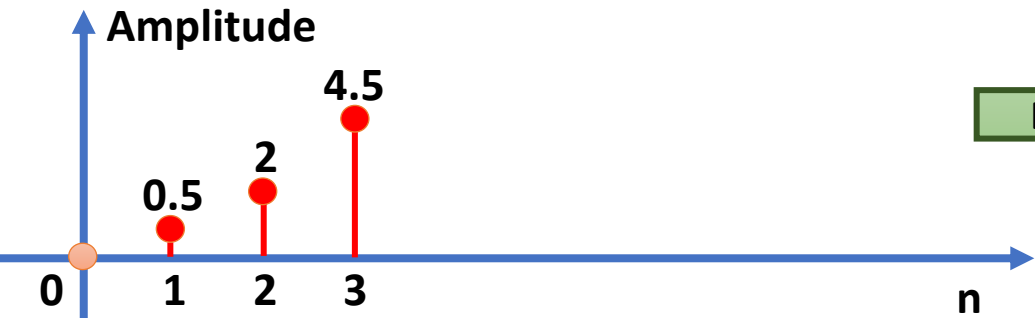


$$\Rightarrow x(-n + 2)$$



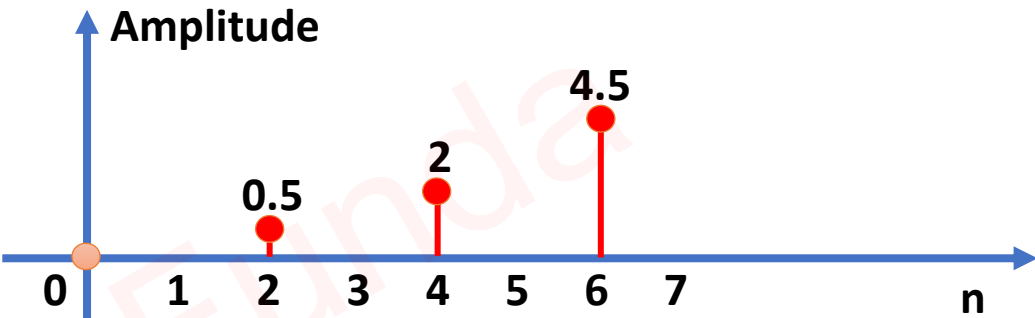
Time Scaling Operation

$\Rightarrow x(n)$

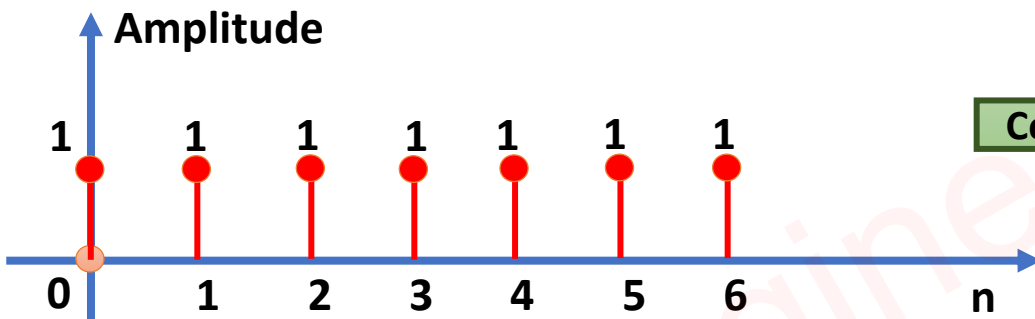


Expansion

$\Rightarrow x(n/2)$

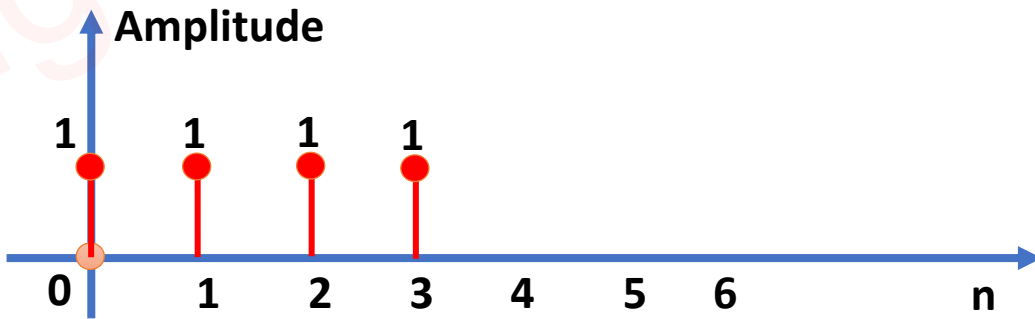


$\Rightarrow x(n)$

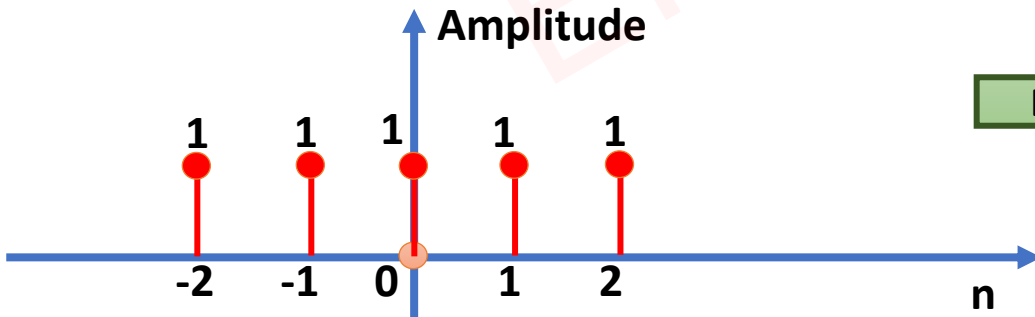


Compression

$\Rightarrow x(2n)$

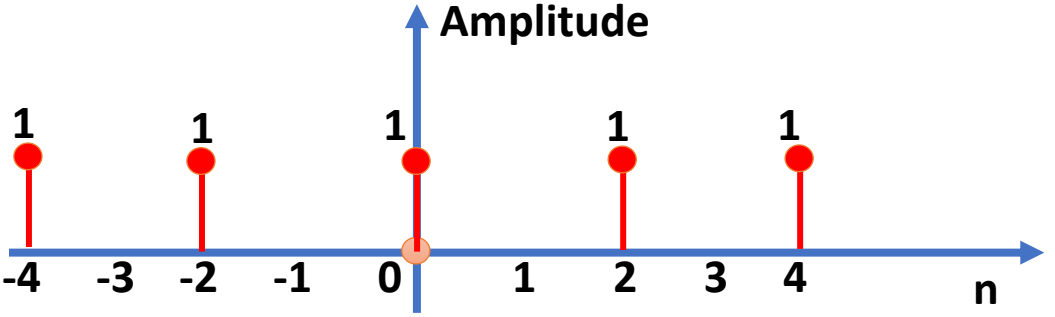


$\Rightarrow x(n)$



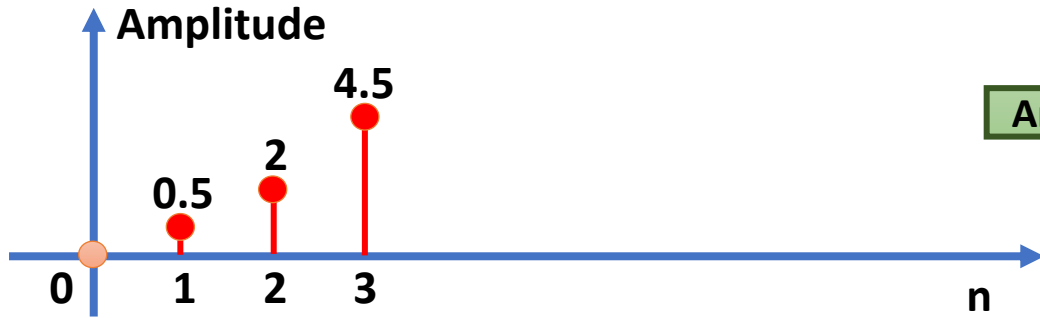
Expansion

$\Rightarrow x(n/2)$



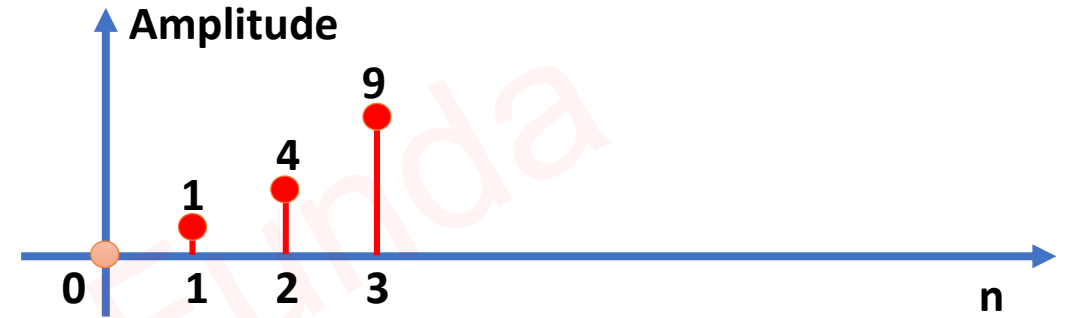
Amplitude Scaling Operation

$\Rightarrow x(n)$

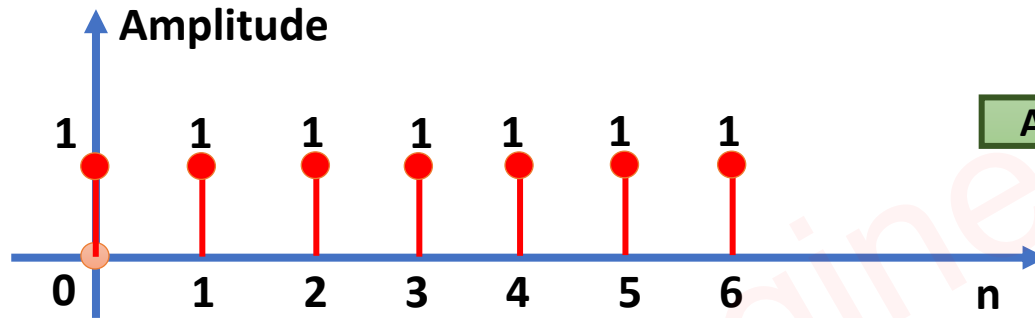


Amplification

$\Rightarrow 2x(n)$

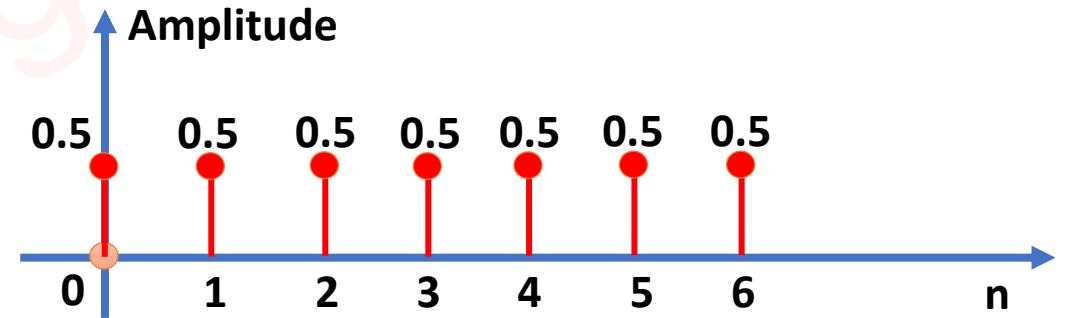


$\Rightarrow x(n)$



Attenuation

$\Rightarrow x(n)/2$



Arithmetic Operation

□ Perform Addition, Subtraction, Multiplication, and Division Operations between $x(n)$ and $y(n)$.

$$\Rightarrow x(n) = \{5, 1, 2, 3, 1\}$$

$$\Rightarrow y(n) = \{0, 1, 2, 0, 2\}$$

$$\Rightarrow \text{Add}(n) = \{5, 2, 4, 3, 3\}$$

$$\Rightarrow \text{Sub}(n) = \{5, 0, 0, 3, -1\}$$

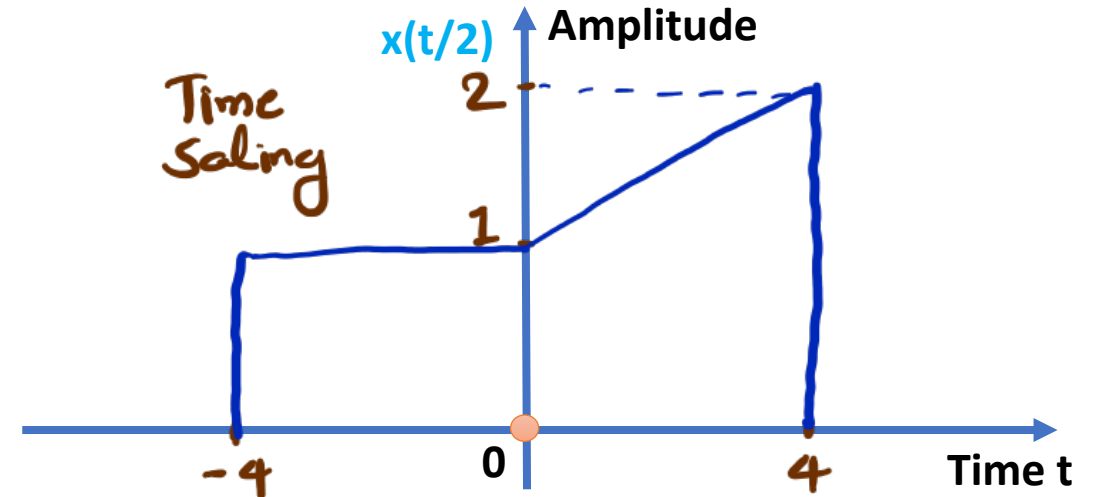
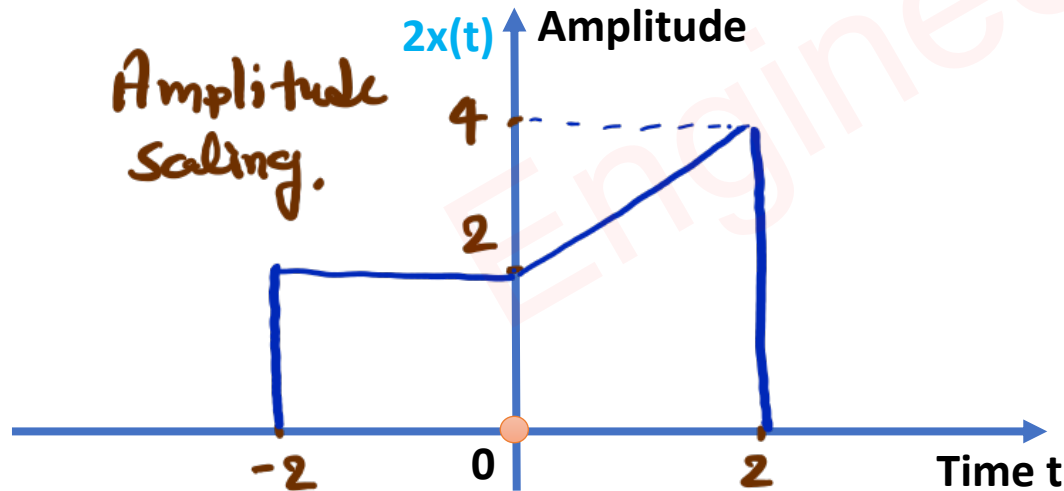
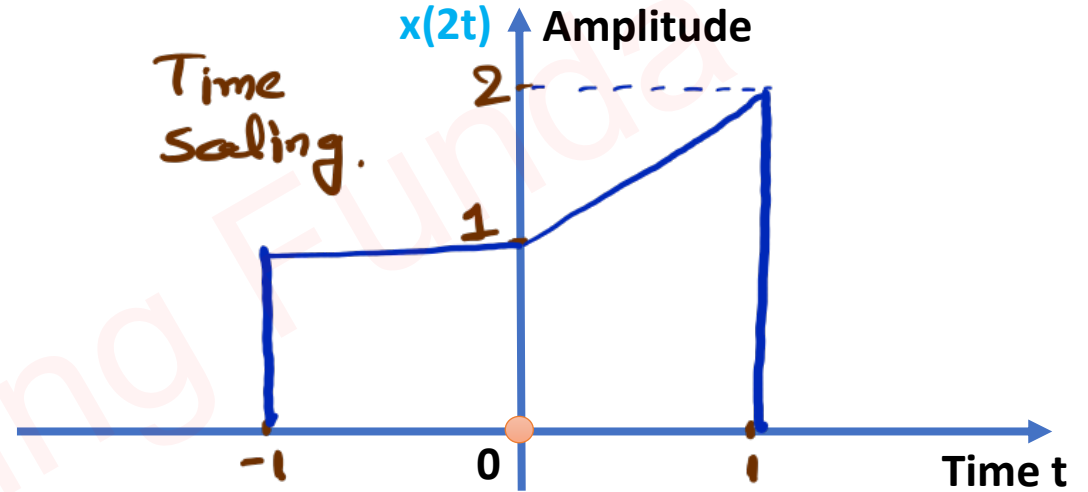
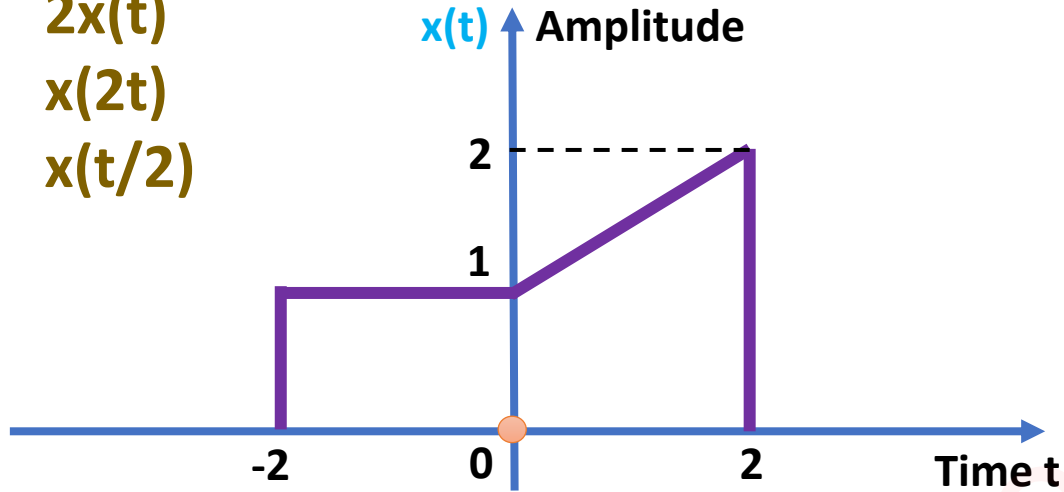
$$\Rightarrow \text{Mul}(n) = \{0, 1, 4, 0, 2\}$$

$$\Rightarrow \text{Div}(n) = \{\infty, 1, 1, \infty, 0.5\}$$

Examples of Operations on Signals

Question 1 – Perform the following Operations on signal $x(t)$

- a. $2x(t)$
- b. $x(2t)$
- c. $x(t/2)$



Question 2 – Perform the following Operations on signal $x(n]$

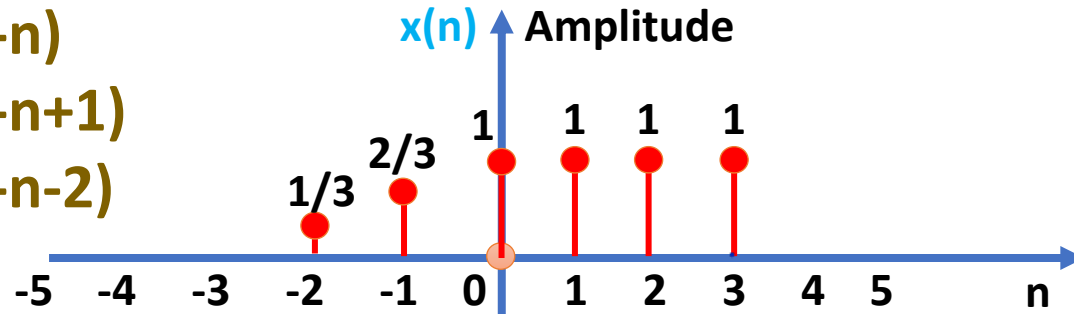
a. $x(n-3]$

b. $x(n+2]$

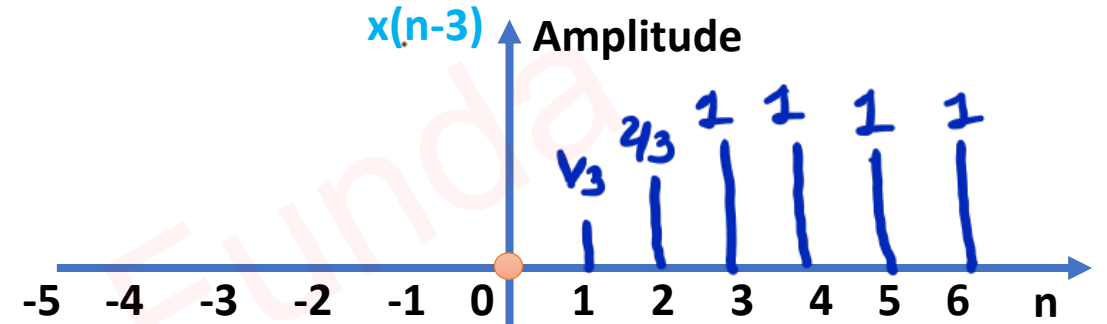
c. $x(-n]$

d. $x(-n+1]$

e. $x(-n-2]$

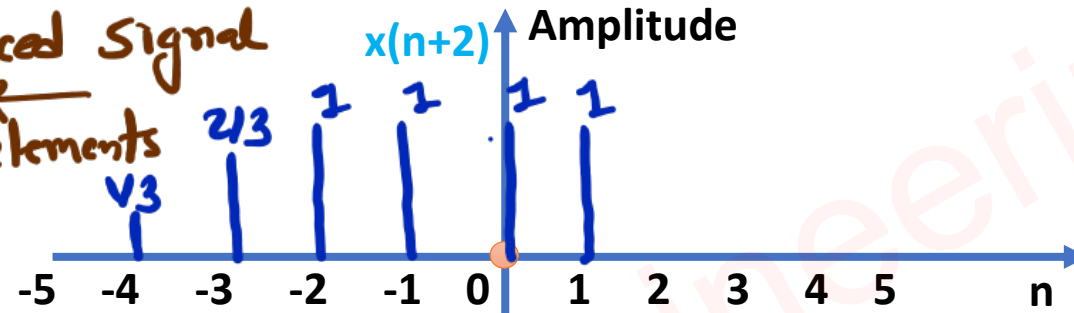


Delayed Signal $\rightarrow$ 3 elements.

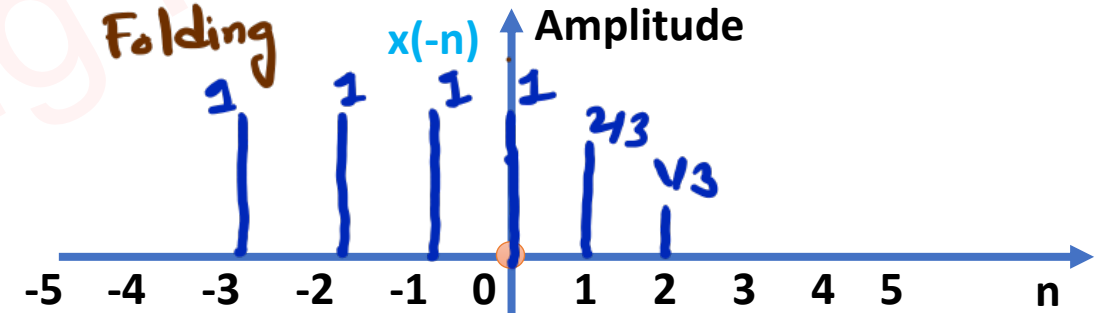


Advanced Signal

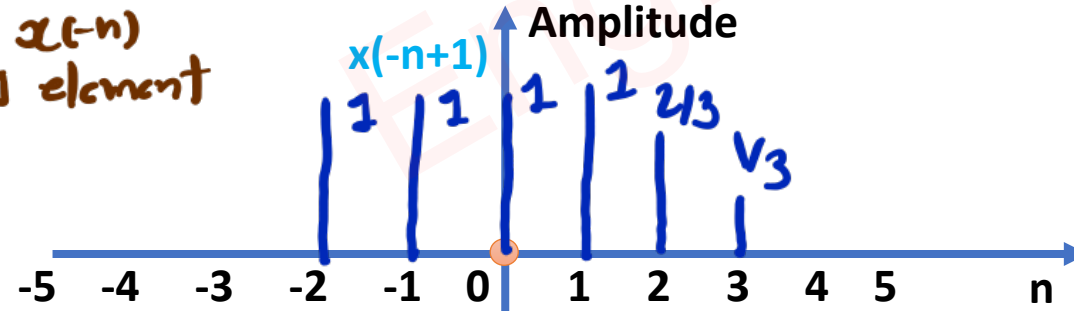
$\leftarrow$
2 elements



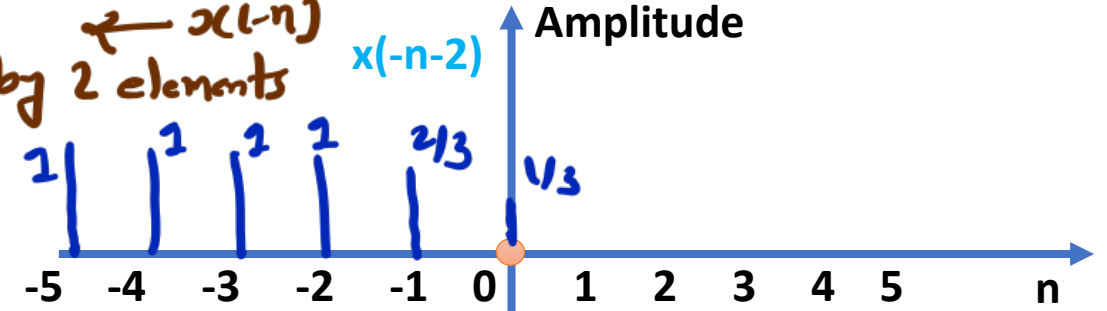
Folding



$\rightarrow$ $x(-n]$
by 1 element



$\leftarrow$ $x(-n]$
by 2 elements



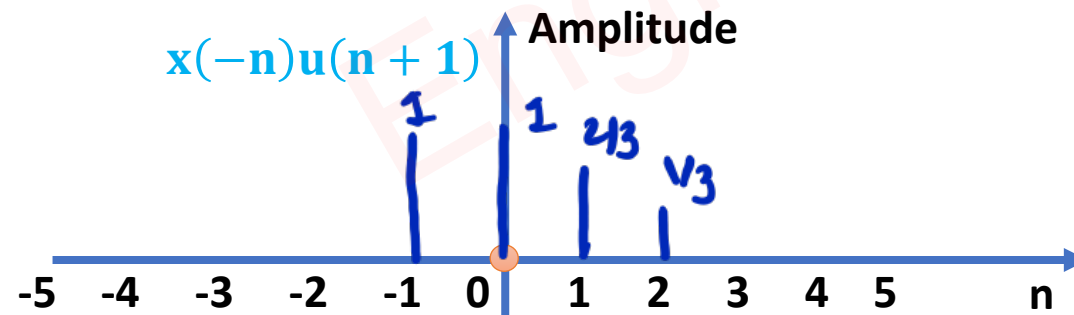
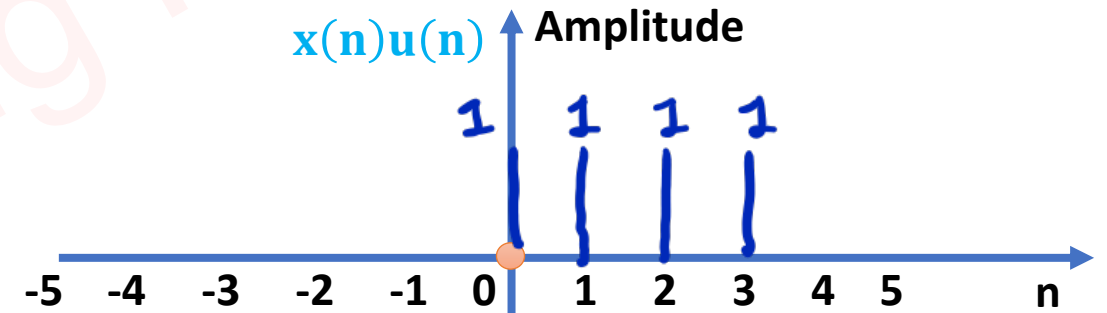
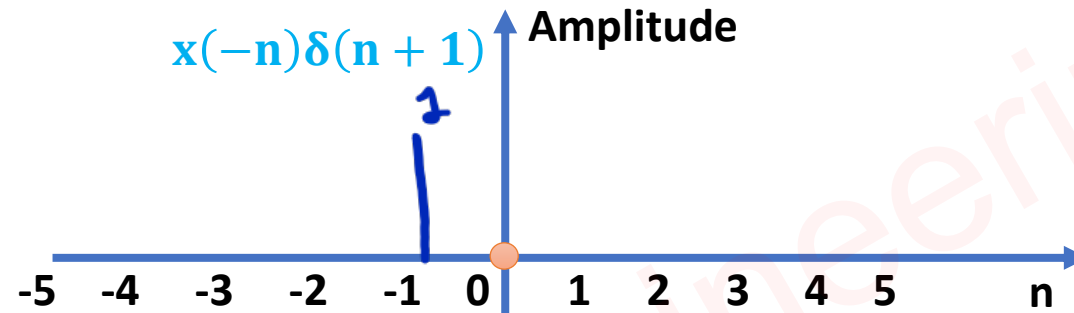
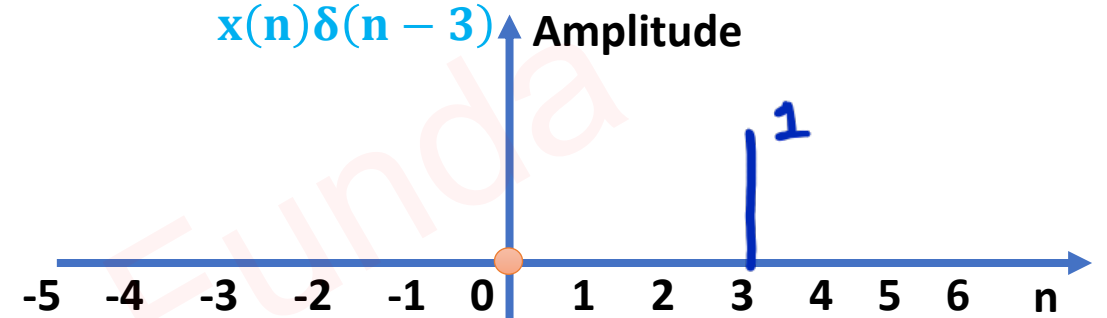
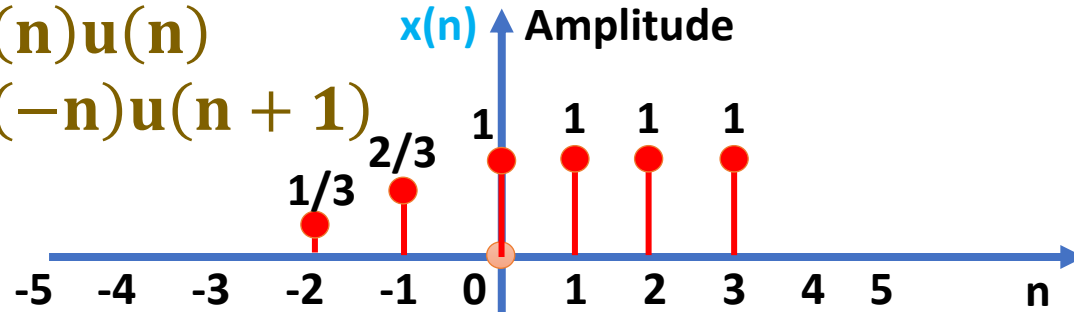
Question 2 – Perform the following Operations on signal $x(n]$

f. $x[n]\delta[n - 3]$

g. $x[-n]\delta[n + 1]$

h. $x[n]u[n]$

i. $x[-n]u[n + 1]$



Question 3 – Perform the following Operations on signal $x(n]$

a. $x(n-3]$

$$\Rightarrow x(n) = \{5, \quad 1, \quad 2, \quad 3, \quad 1\}$$

↑

b. $x(n+2]$

c. $x(-n]$

d. $x(-n+1]$

e. $x(-n-2]$

$$\Rightarrow x(n-3) = \{0, \quad 5, \quad 1, \quad 2, \quad 3, \quad 1, \quad \}$$

↑

$$\Rightarrow x(n+2) = \{5, \quad 1, \quad 2, \quad 3, \quad 1, \quad \}$$

↑

$$\Rightarrow x(-n) = \{1, \quad 3, \quad 2, \quad 1, \quad 5\}$$

↑

$$\Rightarrow x(-n+1) = \{1, \quad 3, \quad 2, \quad 1, \quad 5\}$$

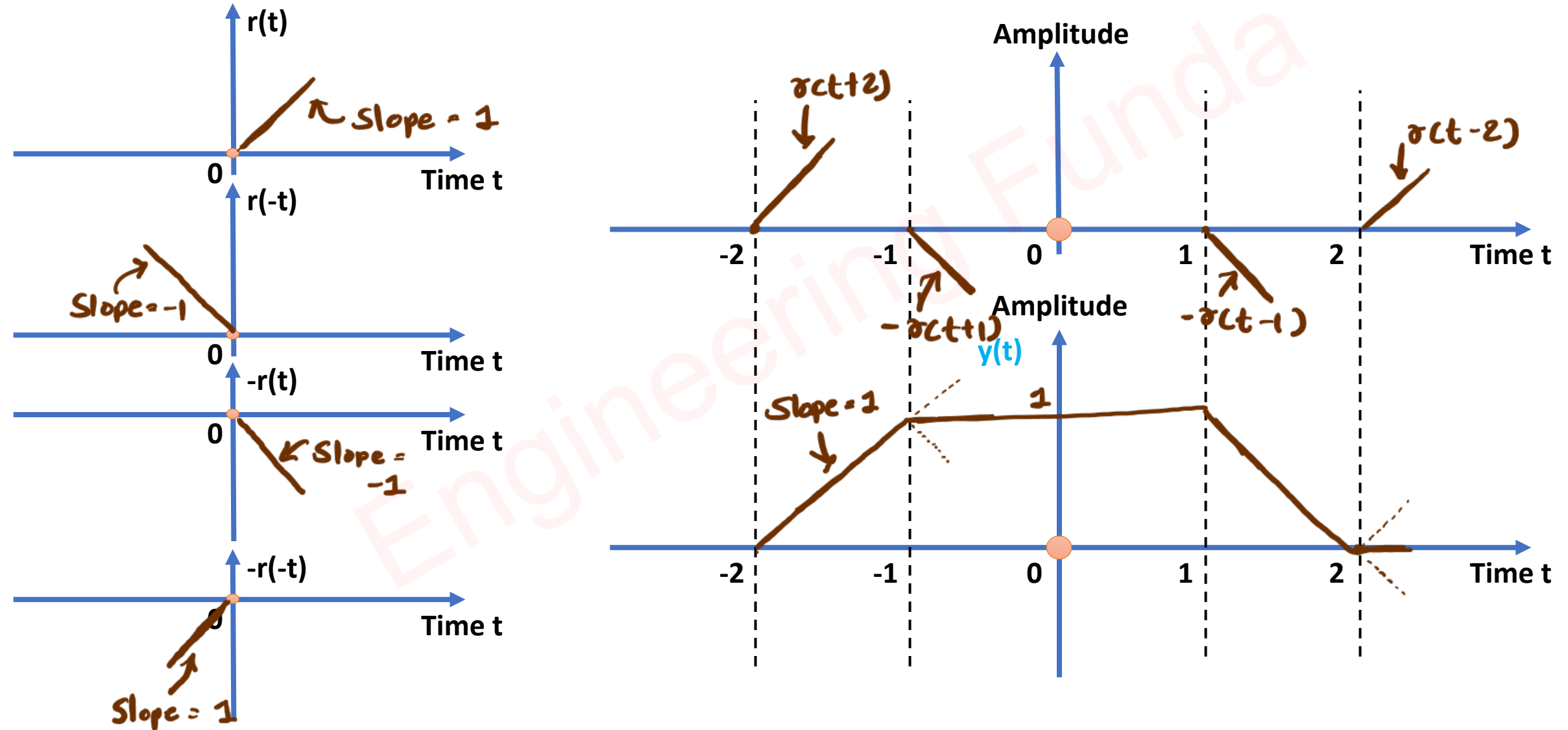
↑

$$\Rightarrow x(-n-2) = \{1, \quad 3, \quad 2, \quad 1, \quad 5\}$$

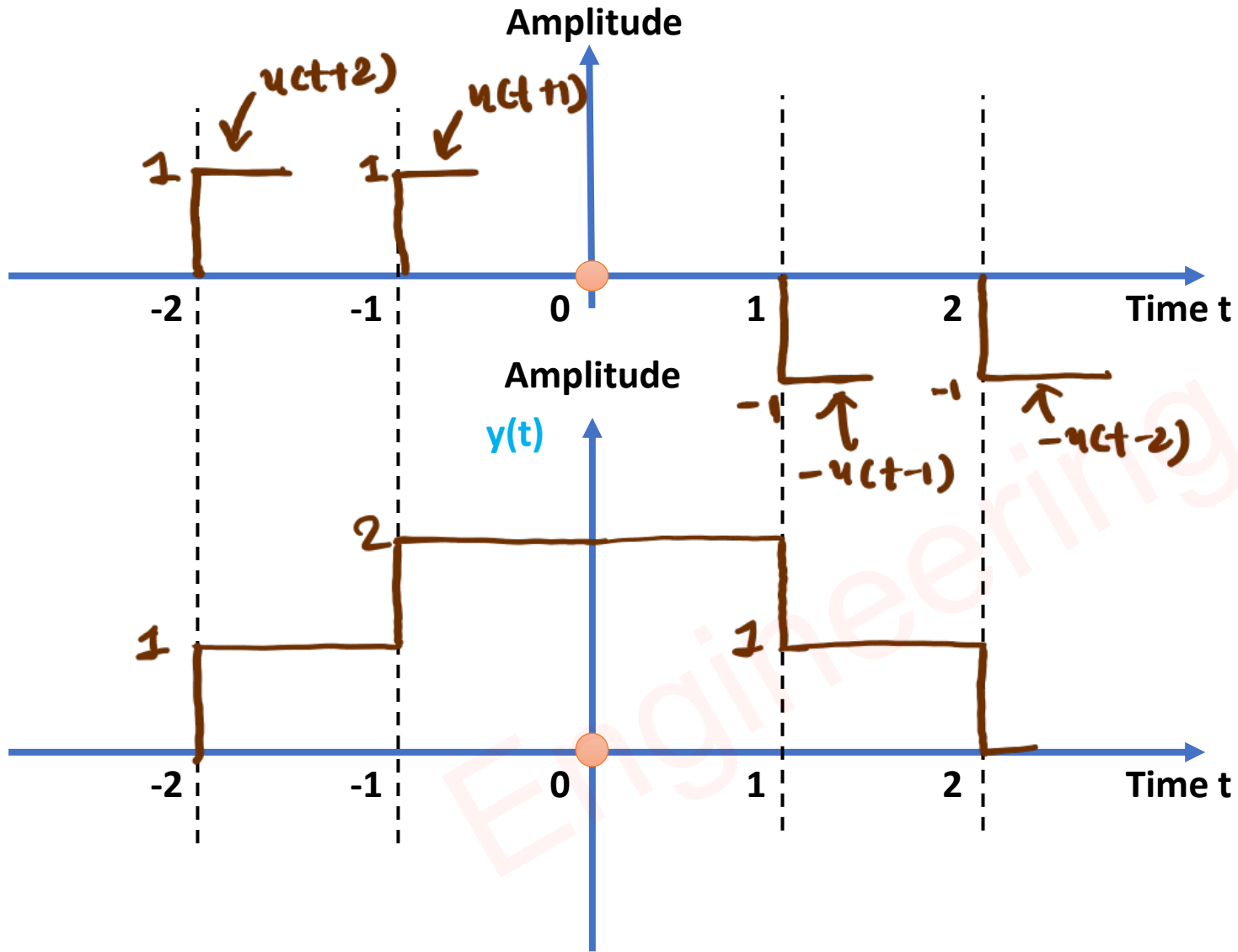
↑

Plot Signal from Function

Question 1 – Draw a signal given by function $y(t) = r(t+2) - r(t+1) - r(t-1) + r(t-2)$.



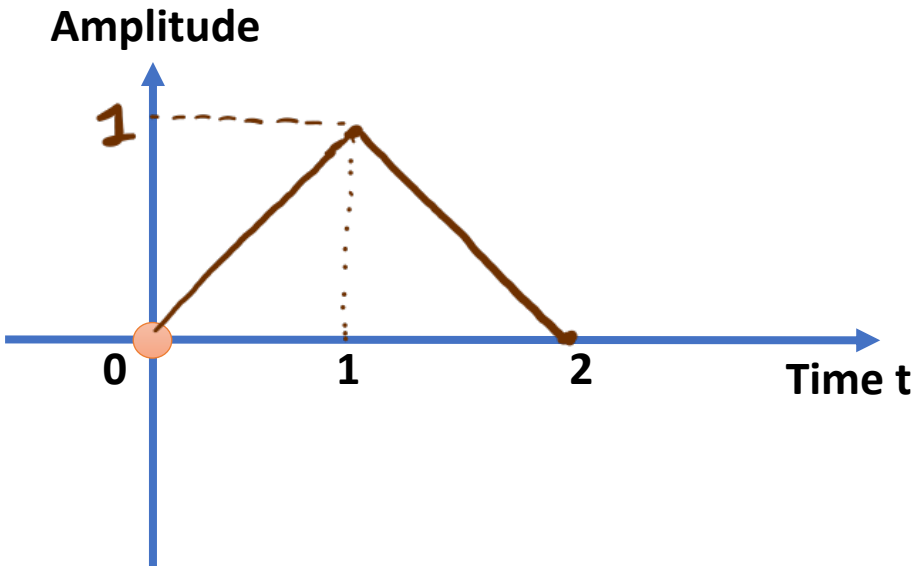
Question 2 – Draw a signal given by function $y(t) = u(t + 2) + u(t + 1) - u(t - 1) - u(t - 2)$.



Question 3 – Draw a signal given by function $y(t)$.

$$\Rightarrow y(t) = t, \quad 0 \leq t \leq 1$$

$$\Rightarrow y(t) = 2 - t, \quad 1 \leq t \leq 2$$



Even and Odd Components of Signal

Basics of Even and Odd Components of Signal $x(t)$:

~> For Signal $x(t)$

$$\Rightarrow x(t) = x_e(t) + x_o(t) \quad \text{--- ①}$$

$$\Rightarrow x(-t) = x_e(t) - x_o(t) \quad \text{--- ②}$$

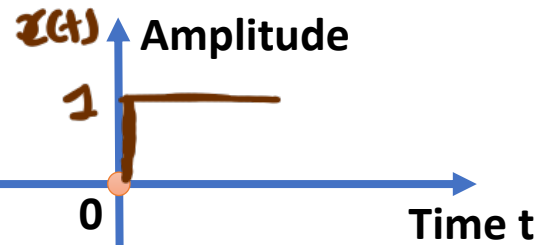
~> Add ① & ②

$$\Rightarrow x_e(t) = \frac{x(t) + x(-t)}{2}$$

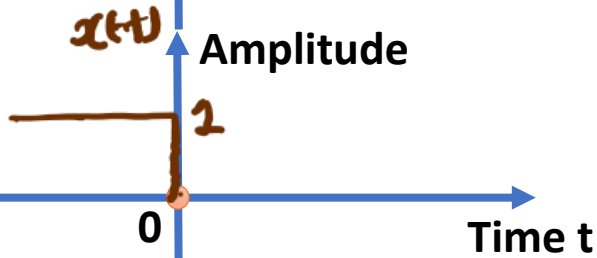
~> Sub ① & ②

$$\Rightarrow x_o(t) = \frac{x(t) - x(-t)}{2}$$

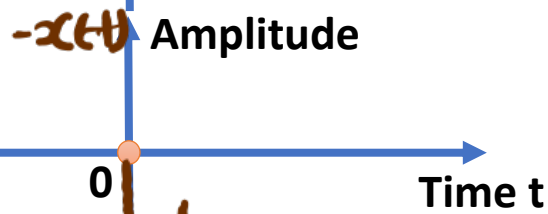
Question 1 – Find Even and Odd Components of Unit Step Signal.



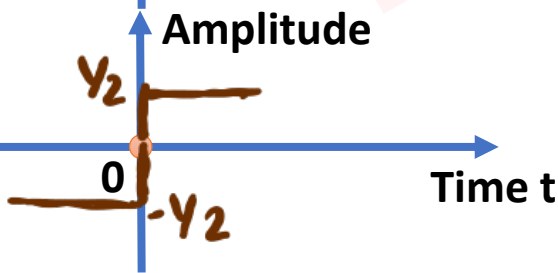
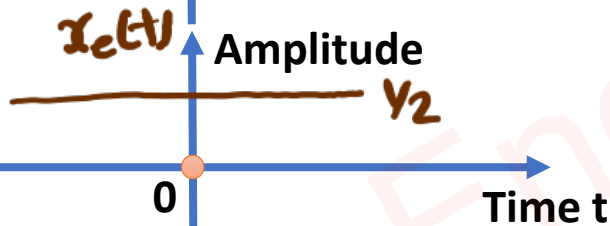
$$\begin{aligned} \leadsto x(t) &= 1 & t \geq 0 \\ &= 0 & t < 0 \end{aligned}$$



$$\leadsto x_e(t) = \frac{x(t) + x(-t)}{2} = \frac{1}{2}$$



$$\leadsto x_o(t) = \frac{x(t) - x(-t)}{2} = \frac{1}{2} \text{sgn}(t)$$



Question 2 – Find Even and Odd Components of $x(t) = \cos t + \sin t + \cos t \sin t + \sin^2 t$.

$$x(-t) = \cos t - \sin t - \cos t \sin t + \sin^2 t$$

$$\Rightarrow x_e(t) = \frac{x(t) + x(-t)}{2} = \cos t + \sin^2 t$$

$$\Rightarrow x_o(t) = \frac{x(t) - x(-t)}{2} = \sin t + \cos t \sin t$$

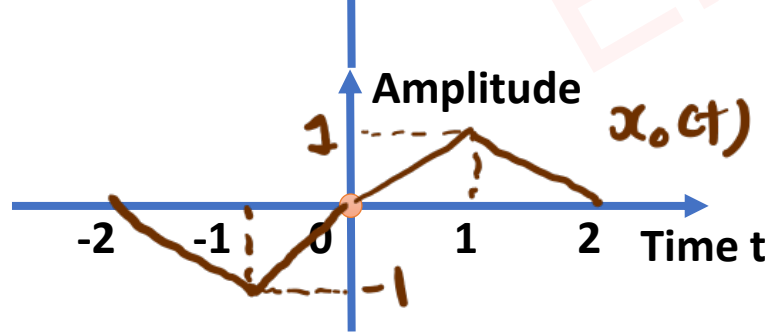
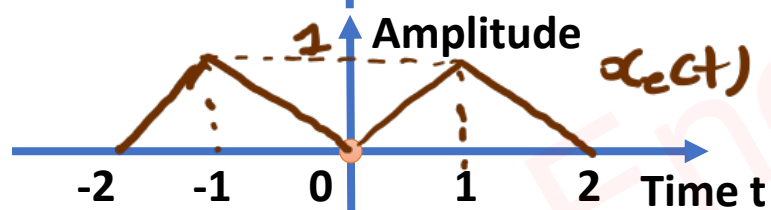
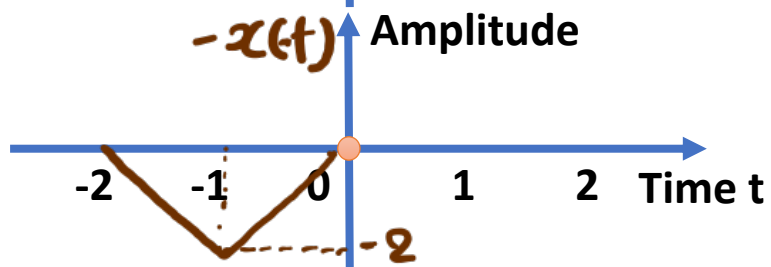
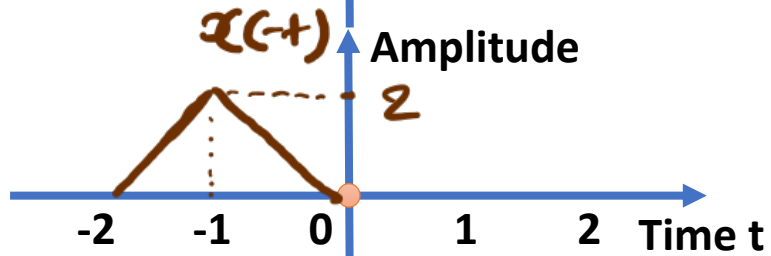
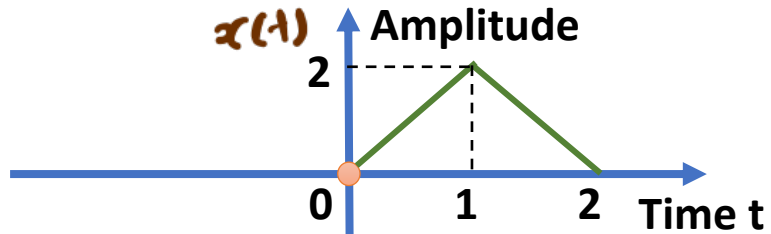
Question 3 – Find Even and Odd Components of $x(t) = 5t^6 + 2t^5 + 3t^4 + 2t^3 + t^2 + 1$.

$$x(-t) = 5t^6 - 2t^5 + 3t^4 - 2t^3 + t^2 + 1$$

$$\leadsto x_e(t) = \frac{x(t) + x(-t)}{2} = 5t^6 + 3t^4 + t^2 + 1$$

$$\leadsto x_o(t) = \frac{x(t) - x(-t)}{2} = 2t^5 + 2t^3$$

Question 4 – Find Even and Odd Components of $x(t)$.



$$- x_e(t) = \frac{x(t) + x(-t)}{2}$$

$$- x_o(t) = \frac{x(t) - x(-t)}{2}$$

Periodicity Check of Signal

Basics of Periodicity of Signal $x(t)$ and $x(n)$:

- ❑ If the Signal Repeats after the finite duration T , then it is said to be a periodic signal.
- ❑ If we have an algebraic addition of two or more periodic signals, then by two different methods, we can identify the periodicity of the signal.
- ❑ **Method-1: Ratio Method**
 - Consider two signals $x_1(t)$ and $x_2(t)$ with periods T_1 and T_2 , respectively. When they are summed, the resultant signal said to be periodic when $T_1/T_2 = \text{Rational Integer Number}$.
- ❑ **Method-2: GCD Method**
 - If GCD of the frequency is possible, then the signal is said to be periodic signal.
- ❑ Fundamental Duration is LCM of Multiple signals.

Question 1 – Find the periodicity of the signal and the fundamental period of

$$x(n) = \cos\left(\underbrace{\frac{6\pi n}{7}}_{\text{Freq.}} + \underbrace{1}_{\text{phase}}\right).$$

$$\rightarrow 2\pi f n = \frac{6\pi n}{7} \Rightarrow f = \frac{3}{7} \left[\frac{\text{cycles}}{\text{Samples}} \right] = \frac{K}{N}$$

→ Fundamental duration

$$N = 7 \text{ samples.}$$

Question 2 – Find the periodicity of the signal and the fundamental period of

$$x(n) = \cos(\pi n^2 / 8).$$

$$\Rightarrow 2\pi f n^2 = \pi n^2 / 8$$

$$\Rightarrow f^2 = \frac{1}{16}$$

$$\Rightarrow f = \frac{1}{4} \left[\frac{\text{cycles}}{\text{Samples}} \right] = \frac{K}{N}$$

→ Fundamental period

$$\underline{\underline{N = 4 \text{ samples}}}$$

Question 3 – Find the periodicity of signal $x(t) = \cos 5t + \sin \pi t$.

$$\Rightarrow \omega_1 = 5$$

$$\Rightarrow T_1 = \frac{2\pi}{\omega_1}$$

$$= \frac{2\pi}{5}$$

$$\Rightarrow \omega_2 = \pi$$

$$\Rightarrow T_2 = \frac{2\pi}{\omega_2}$$

$$= \frac{2\pi}{\pi} = 2$$

$$\rightarrow \frac{T_1}{T_2} = \frac{2\pi/5}{2} = \frac{\pi}{5}$$

↑

It is not Rational Integer number.

→ So, $x(t)$ is not periodic signal.

Question 4 – Find the periodicity of the signal and the fundamental period of $x(t) = \cos \pi t + \sin 4\pi t$.

$$\omega_1 = \pi$$

$$\omega_2 = 4\pi$$

$$\Rightarrow T_1 = \frac{2\pi}{\omega_1} = 2$$

$$\Rightarrow T_2 = \frac{2\pi}{\omega_2} = \frac{1}{2}$$

$$\rightarrow \frac{T_1}{T_2} = \frac{2}{1/2} = 4 \leftarrow \text{It is rational Integer Number.}$$

$\rightarrow$ So, $x(t)$ is periodic signal.

$\rightarrow$ Fundamental period $T = \text{LCM of } T_1 \text{ \& } T_2$
 $= 2 \text{ sec.}$

Question 5 – Find the periodicity of the signal and the fundamental period of $x(t) = \cos \pi t + \cos 3\pi t + \sin 5\pi t$.

$$\omega_1 = \pi$$

$$\omega_2 = 3\pi$$

$$\omega_3 = 5\pi$$

$$\Rightarrow T_1 = \frac{2\pi}{\omega_1} \\ = 2 \text{ Sec}$$

$$\Rightarrow T_2 = \frac{2\pi}{\omega_2} \\ = \frac{2}{3} \text{ Sec}$$

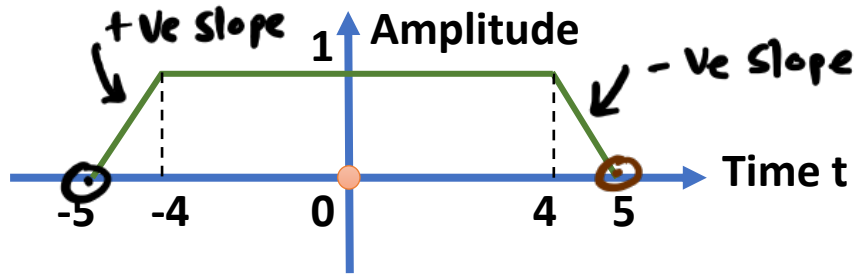
$$\Rightarrow T_3 = \frac{2\pi}{\omega_3} \\ = \frac{2}{5} \text{ Sec}$$

$$\Rightarrow T_1 : T_2 : T_3 = 2 : \frac{2}{3} : \frac{2}{5} \\ = 30 : 10 : 6 \\ = (15 : 5 : 3) \underline{\underline{2}}$$

$$\Rightarrow \text{Fundamental duration} = \text{LCM of } T_1, T_2 \text{ \& } T_3 \\ = 2 \text{ Sec}$$

Energy & Power of Signal From Waveforms

Question 1 – Find the Energy and Power of the given Signal $x(t)$.



~ For $-5 \leq t \leq -4$

$$\begin{aligned}x(t) &= mt + C \\&= t + C\end{aligned}$$

$$\sim (t, x(t)) = (-5, 0)$$

$$\Rightarrow 0 = -5 + C$$

$$\Rightarrow C = 5$$

$$\sim x(t) = t + 5$$

~ For $-4 \leq t \leq 4$

$$- x(t) = 1$$

~ For $4 \leq t \leq 5$

$$\begin{aligned}- x(t) &= mt + C \\&= -t + C\end{aligned}$$

$$\sim (t, x(t)) = (5, 0)$$

$$\Rightarrow 0 = -5 + C$$

$$\Rightarrow C = 5$$

$$- x(t) = -t + 5$$

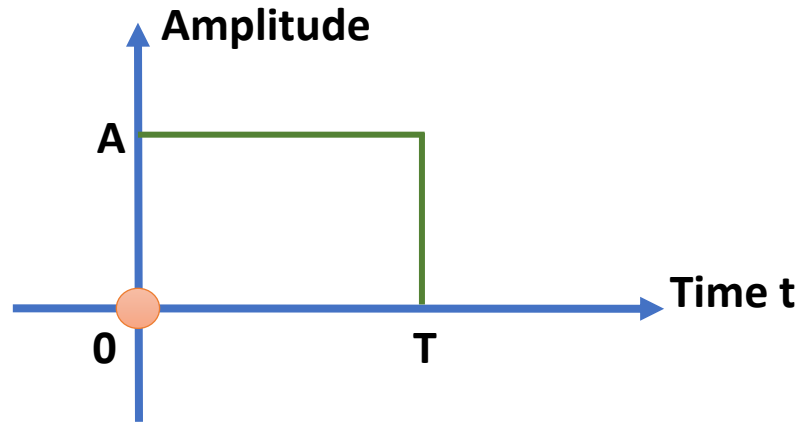
$$\begin{aligned} \leadsto x(t) &= t+5 & -5 \leq t \leq -4 \\ &= 1 & -4 \leq t \leq 4 \\ &= -t+5 & 4 \leq t \leq 5 \end{aligned}$$

$$\begin{aligned} \leadsto E &= \int_{-\infty}^{\infty} |x(t)|^2 dt \\ &= \int_{-5}^{-4} (t+5)^2 dt + \int_{-4}^4 1 dt + \int_4^5 (-t+5)^2 dt \\ &= \int_{-5}^{-4} (t^2 + 10t + 25) dt + \int_{-4}^4 1 dt + \int_4^5 (t^2 - 10t + 25) dt \\ &= \left[\frac{t^3}{3} + 5t^2 + 25t \right]_{-5}^{-4} + [t]_{-4}^4 + \left[\frac{t^3}{3} - 5t^2 + 25t \right]_4^5 \end{aligned}$$

$$E = 8.66 \text{ J}$$

$\leadsto$ For energy signal, $P = 0$.

Question 2 – Find the Energy and Power of the given Pulse Signal $x(t)$.

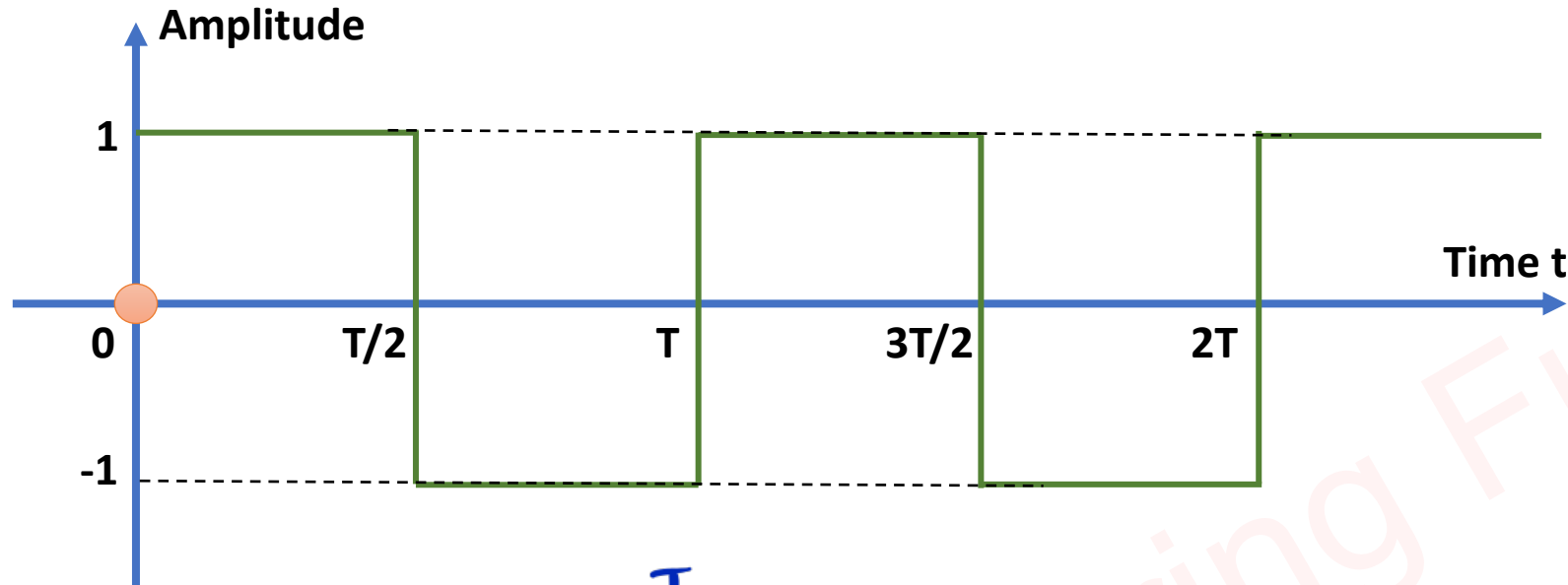


$$x(t) = A, \quad 0 < t < T$$
$$= 0, \quad \text{elsewhere}$$

$$\begin{aligned} \leadsto E &= \int_{-\infty}^{\infty} |x(t)|^2 dt \\ &= \int_0^T A^2 dt \\ &= A^2 [t]_0^T \\ &= A^2 T \end{aligned}$$

$\leadsto$ For Energy Signal, $P=0$

Question 3 – Find the Energy and Power of the given Square wave Signal $x(t)$.



⇒ Periodic signal.

$$x(t) = 1, \quad 0 < t < T/2 \\ = -1, \quad T/2 < t < T$$

$$\sim P = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T x^2(t) dt$$

$$= \lim_{T \rightarrow \infty} \frac{1}{T} \left[\int_0^{T/2} 1^2 dt + \int_{T/2}^T 1^2 dt \right]$$

$$= \lim_{T \rightarrow \infty} \frac{1}{T} [T]$$

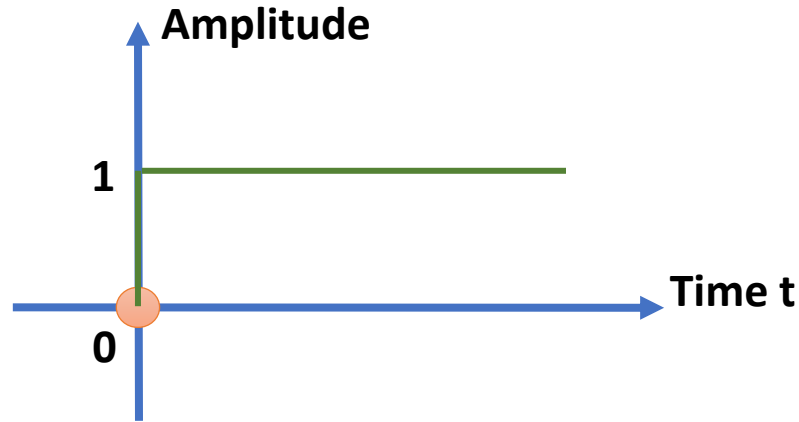
$$= 1 \text{ W}$$

⇒ For power signal,

$$E = \infty$$

Energy & Power of Step & Exponential Signal

Question 1 – Find the Energy and Power of the Unit Step Signal $x(t)$.



⇒ Unit Step Signal

$$u(t) = 1, \quad t \geq 0 \\ = 0, \quad t < 0$$

⇒ Energy E .

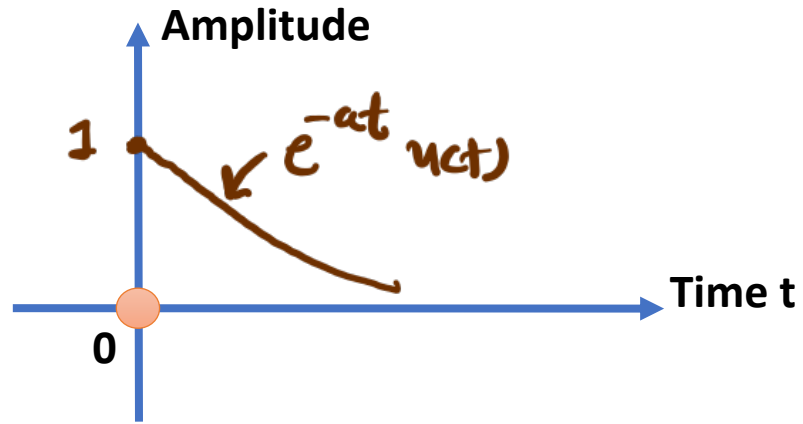
$$\begin{aligned} - E &= \int_{-\infty}^{\infty} |x(t)|^2 dt \\ &= \int_0^{\infty} 1 dt \\ &= [t]_0^{\infty} \\ &= \infty \end{aligned}$$

⇒ Power P

$$\begin{aligned} - P &= \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T |x(t)|^2 dt \\ &= \lim_{T \rightarrow \infty} \frac{1}{2T} \int_0^T 1 dt \\ &= \lim_{T \rightarrow \infty} \frac{1}{2T} (T) \\ &= \frac{1}{2} \end{aligned}$$

⇒ Step Signal is Power Signal.

Question 2 – Find the Energy and Power of the Exponential Signal $e^{-at}u(t)$.



Signal $x(t)$

$$x(t) = e^{-at} \quad t \geq 0$$
$$= 0 \quad t < 0$$

Energy E

$$\Rightarrow E = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

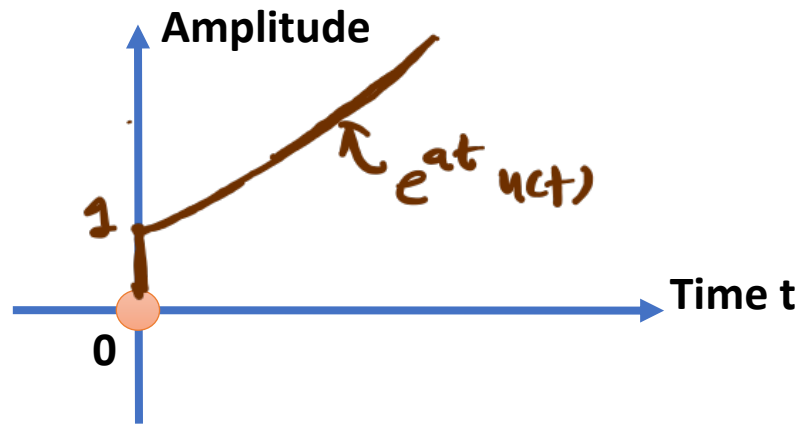
$$= \int_0^{\infty} e^{-2at} dt$$

$$= \left[\frac{e^{-2at}}{-2a} \right]_0^{\infty}$$

$$= -\frac{1}{2a} [e^{-\infty} - e^0] = \frac{1}{2a}$$

For energy signal, power $P = 0$

Question 3 – Find the Energy and Power of the Exponential Signal $e^{at}u(t)$.



~ Signal $x(t)$

$$x(t) = e^{at}, \quad t \geq 0$$
$$= 0, \quad t < 0$$

Hence, given signal is not energy & power signal.

~ Energy E

$$\Rightarrow E = \int_{-\infty}^{\infty} |x(t)|^2 dt$$
$$= \int_0^{\infty} e^{2at} dt$$
$$= \left[\frac{e^{2at}}{2a} \right]_0^{\infty}$$
$$= \frac{1}{2a} [e^{\infty} - e^0]$$
$$= \infty$$

~ Power P

$$\Rightarrow P = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T |x(t)|^2 dt$$

$$= \lim_{T \rightarrow \infty} \frac{1}{2T} \int_0^T e^{2at} dt$$

$$= \lim_{T \rightarrow \infty} \frac{1}{2T} \left[\frac{e^{2at}}{2a} \right]_0^T$$

$$= \lim_{T \rightarrow \infty} \frac{1}{2T} \left[\frac{e^{2aT} - 1}{2a} \right] = 0$$

Energy & Power of Discrete Signal

Basics of Energy & Power of Discrete Signal

□ Energy of Discrete Signal $x(n)$ is given by:

$$\Rightarrow E = \sum_{n=-\infty}^{n=\infty} |x(n)|^2$$

□ Power of Discrete Signal $x(n)$ is given by:

$$\Rightarrow P = \lim_{N \rightarrow \infty} \frac{1}{2N + 1} \sum_{n=-N}^{n=N} |x(n)|^2$$

□ All Finite Time and Amplitude Signals $x(n)$ are Energy Signals with **ZERO Power**.

□ All Periodic Signals $x(n)$ are Power Signals with **INFINITE Energy**.

Question 1 – Find the Energy and Power of the given Signal $x(n) = e^{j4n}$.

$$\begin{aligned}\leadsto x(n) &= e^{j4n} \\ &= \cos 4n + j \sin 4n\end{aligned}$$

$\leadsto$ Power of $x(n)$

$$\Rightarrow P = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x(n)|^2$$

$$= \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N 1$$

$$= \lim_{N \rightarrow \infty} \frac{1}{\cancel{2N+1}} (\cancel{2N+1})$$

$$= 1 \text{ W}$$

$\leadsto$ For Power Signal, $E = \infty$

Question 2 – Find the Energy and Power of the given Signal $x(n)$.

$$\Rightarrow x(n) = \left(\frac{1}{2}\right)^n, \quad n \geq 0$$

$$\Rightarrow x(n) = (3)^n, \quad n < 0$$

~ Energy of $x(n)$

$$\Rightarrow E = \sum_{n=-\infty}^{\infty} |x(n)|^2$$

$$= \sum_{n=0}^{\infty} \left(\frac{1}{4}\right)^n + \sum_{n=-1}^{-\infty} (1)^n$$

$$= \sum_{n=0}^{\infty} \left(\frac{1}{4}\right)^n + \sum_{n=1}^{\infty} \left(\frac{1}{4}\right)^n$$

$$\sum_{n=0}^{\infty} a^n = \frac{1}{1-a} \qquad \sum_{n=1}^{\infty} a^n = \frac{a}{1-a}$$

$$= \frac{1}{1-1/4} + \frac{1/4}{1-1/4} = \frac{4}{3} + \frac{1}{8} = \frac{35}{24} \text{ J}$$

~ For Energy signal, $P=0$

Energy & Power of Signal

Question 1 – Find the Energy and Power of the Signal $y(t)$. Where, $x(t) = r(t) - r(t - 2)$ and $y(t) = \frac{dx(t)}{dt}$.

$$\Rightarrow y(t) = \frac{dx(t)}{dt} = \frac{d(r(t) - r(t-2))}{dt} = u(t) - u(t-2)$$

$$\Rightarrow y(t) = 1, \quad t \in (0, 2)$$

$\Rightarrow$ Energy E

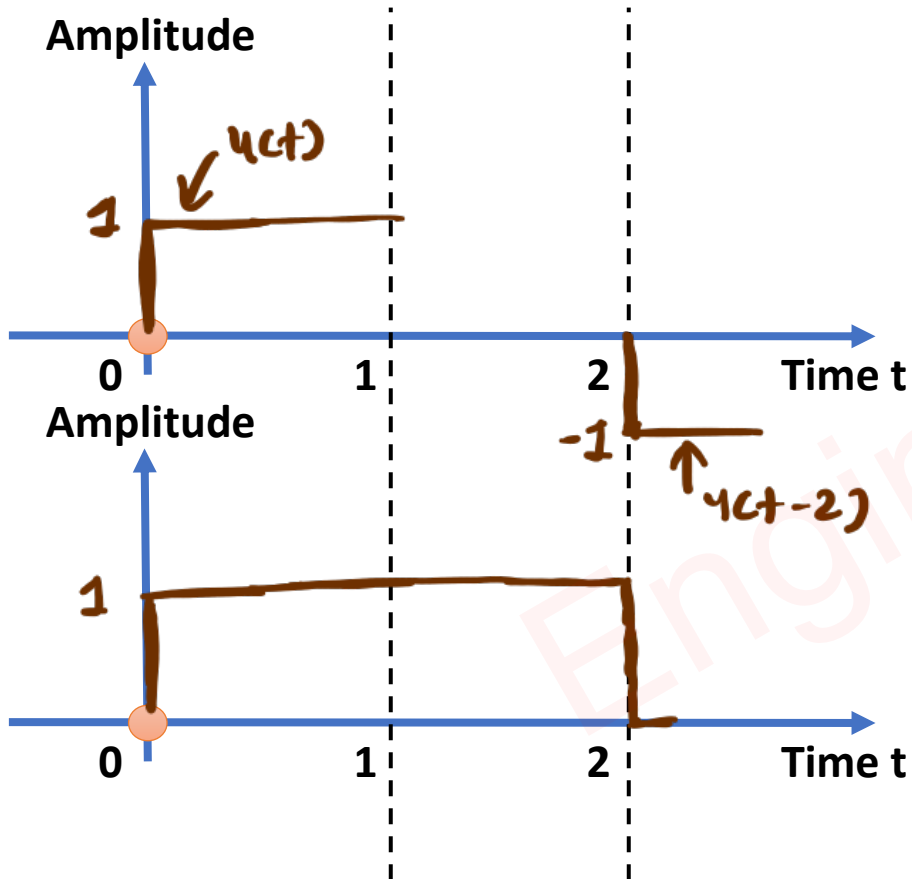
$$E = \int_{-\infty}^{\infty} |y(t)|^2 dt$$

$$= \int_0^2 1^2 dt$$

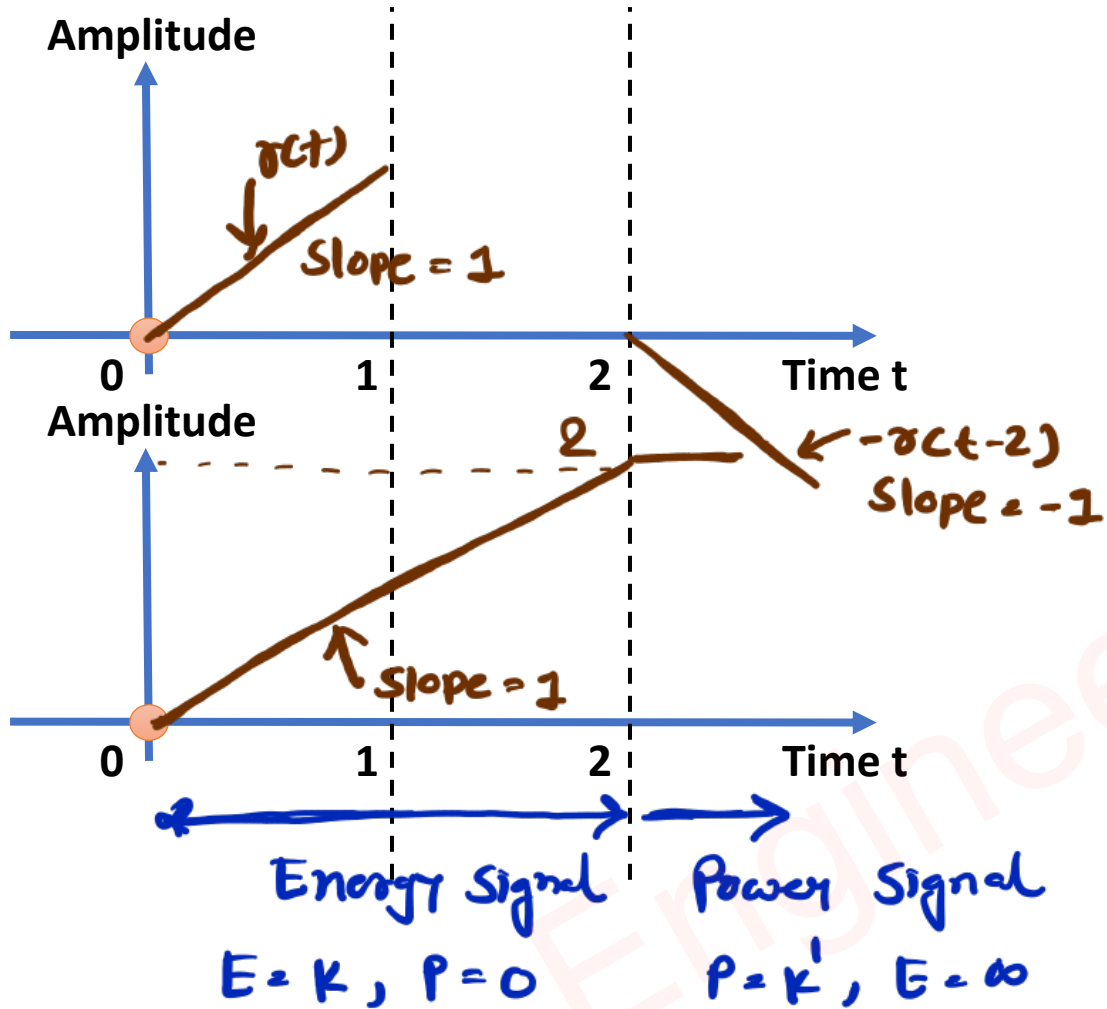
$$= [t]_0^2$$

$$= 2 \text{ J}$$

$\Rightarrow$ For Energy Signal,
 $P = 0$.



Question 2 – Find the Energy and Power of the given Signal $x(t) = r(t) - r(t - 2)$.



∴ Hence, total signal is power signal.

∴ Power P

$$\Rightarrow P = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T |x(t)|^2 dt$$

$$= \lim_{T \rightarrow \infty} \frac{1}{2T} \int_2^T 2^2 dt$$

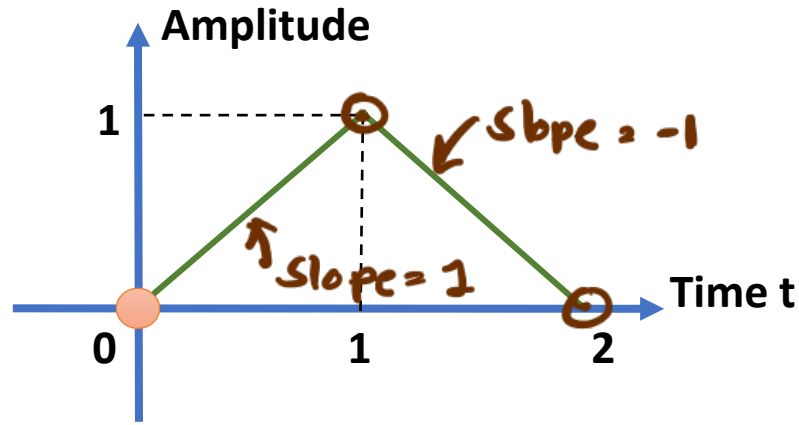
$$= \lim_{T \rightarrow \infty} \frac{1}{2T} [4(T-2)]$$

$$= \lim_{T \rightarrow \infty} 2 - \frac{4}{T}$$

$$= \underline{\underline{2 \text{ W}}}$$

∴ For power signal, $E = \infty$

Question 3 – Find the Energy and Power of the given Signal.



$$\leadsto x(t) \rightarrow t \in (0,1) \Rightarrow x(t) = t$$

$$\leadsto x(t) \rightarrow t \in (1,2) \Rightarrow x(t) = mt + c \\ = -t + c$$

$$\leadsto \text{Put } (t, x(t)) = (2, 0)$$

$$\Rightarrow 0 = -2 + c \Rightarrow c = 2$$

$$\leadsto x(t) = t, \quad t \in (0,1) \\ x(t) = -t + 2, \quad t \in (1,2)$$

$\leadsto$ Energy E

$$\Rightarrow E = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

$$= \int_0^1 t^2 dt + \int_1^2 (t^2 - 4t + 4) dt$$

$$= \left[\frac{t^3}{3} \right]_0^1 + \left[\frac{t^3}{3} - 2t^2 + 4t \right]_1^2 = \frac{2}{3}$$

$\leadsto$ For Energy Signal, $P=0$

Energy of Time Scaled Signal & Verification by Parseval's Theorem

Question 1 – A signal $x(t)$ has Energy E , Calculate the energy of $x(3t)$.

⇒ Energy for $x(t)$ is given by:

$$\rightarrow E' = E/3$$

$$\Rightarrow E = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

⇒ Energy for $x(at)$ is given by:

$$\Rightarrow E' = \int_{-\infty}^{\infty} |x(at)|^2 dt$$

$$\begin{aligned} \text{Let, } at = t' &\Rightarrow a dt = dt' \\ &\Rightarrow dt = dt'/a \end{aligned}$$

$$- t' \in (-\infty, \infty)$$

$$\Rightarrow E' = \int_{-\infty}^{\infty} |x(t')|^2 dt'/a = \frac{1}{a} E$$

Question 2 – Find the Energy and verify by Parseval's Theorem for the signal $x(t) = e^{-4t}u(t)$.

$$\begin{aligned} \Rightarrow x(t) &= e^{-4t} & t \geq 0 \\ &= 0 & t < 0 \end{aligned}$$

Energy of $x(t)$ is given by:

$$\Rightarrow E = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

$$= \int_0^{\infty} (e^{-4t})^2 dt$$

$$= \int_0^{\infty} e^{-8t} dt$$

$$= \left[\frac{e^{-8t}}{-8} \right]_0^{\infty}$$

$$= -\frac{1}{8} [e^{-\infty} - e^0]$$

$$= 1/8$$

Energy of $x(\omega)$ by parseval's theorem

$$\Rightarrow E = \frac{1}{2\pi} \int_{-\infty}^{\infty} |x(\omega)|^2 d\omega$$

$$- x(t) = e^{-4t}u(t) \xrightarrow{LT} x(\omega) = \frac{1}{4+j\omega}$$

$$\Rightarrow |x(\omega)|^2 = \frac{1}{16+\omega^2}$$

$$\Rightarrow E = \frac{1}{2\pi} \int_{-\infty}^{\infty} \left(\frac{1}{16+\omega^2} \right) d\omega$$

$$\left[\int \frac{a}{a^2+\omega^2} d\omega = \tan^{-1}\left(\frac{\omega}{a}\right) \right]$$

$$\Rightarrow E = \frac{1}{8\pi} \int_{-\infty}^{\infty} \left(\frac{4}{16+\omega^2} \right) d\omega$$

$$= \frac{1}{8\pi} \left[\tan^{-1} \left(\frac{\omega}{4} \right) \right]_{-\infty}^{\infty}$$

$$= \frac{1}{8\pi} \left[\pi/2 - (-\pi/2) \right]$$

$$= 1/8$$